802 **Poster Session**

Evaluation of social support, discrimination, and perceived health in patients with gastrointestinal tract cancers.

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Background: Health-related social conditions affect cancer outcomes. Gastrointestinal cancers (GIC) comprise a large proportion of cancers diagnosed in the United States. We hypothesized that a diagnosis of GIC negatively impacts Social Determinants of Health (SDOH) and perceived health (PH). **Methods:** Prospectively collected data from The All of Us database was queried for GIC patients who completed SDOH and PH surveys. Healthy, non-cancer (NC) participants were matched to GIC based on age, race, sex, deprivation index, and Charlson Deyo comorbidities using Propensity Score Matching. Regression was used to analyze the relationship between cancer status and survey scores. **Results:** A total of 231,033 patients met the criteria: 14,032 GIC and 217,001 NC. The majority of patients were white (69%%), female (59.2%), and non-Hispanic (97.2%). Colorectal cancer was most represented (81%), followed by anus (12%), stomach (3%), and pancreatic (1%) cancers. The median time between cancer diagnosis and survey response was 3.4 years (IQR 1.4-6.5 years). With respect to SDOH, GIC patients perceived more social support (β 0.07 [95% CI: 0.04, 0.10]), more neighborhood positivity (β 0.07 [95% CI: 0.05, 0.10]) and less healthcare discrimination (β -0.03 [95% CI: -0.06, -0.01]) (Table, NC is reference). GIC surveyed < 1 year after their diagnosis scored higher on social support than those surveyed > 1 year (β 0.13 [95% CI: 0.05, 0.21]). For PH, GIC were 15% more likely to report better mental health and 33% more likely to report better quality of life (QOL). Those diagnosed < 1 year before their survey had no difference in perceived physical and general health compared to NC, although their QOL (OR 1.3 [95% CI: 1.2-1.4]) and mental health (OR 1.1 [95% CI: 1.0-1.2]) were similar to those surveyed > 1 year after the diagnosis. Chemotherapy-treated GIC patients reported better PH in all four domains than those not treated with chemotherapy, including QOL (OR 1.3, [95% CI: 1.2-1.5]) and mental health (OR 1.2, [95% CI: 1.1-1.4]). **Conclusions:** Findings of this survey study reveal that a GI cancer diagnosis has a favorable impact on SDOH and PH - GIC feel more supported, less discriminated, and perceive better health compared to NC, and time of cancer diagnosis and chemotherapy impact outcomes. A concerted societal effort is required to improve health related social conditions for individuals with and without cancer. Research Sponsor: None.

Impact of cancer status on perceived health and social determinants of health.

Social Determinants of Health*	Beta (95% CI)	P Value
Social Support	0.074 (0.04, 0.10)	< 0.001
Neighborhood Positive	0.074 (0.05, 0.10)	< 0.001
Healthcare Discrimination	-0.034 (-0.06, -0.01)	< 0.001
UCLA Loneliness	-0.013 (-0.03, 0.01)	0.196
Perceived Health Domain*	**Odds Ratio (95% CI)**	**P Value**
Quality of Life	1.32 (1.27, 1.38)	< 0.001
Physical Health	1.27 (1.21, 1.32)	< 0.001
Mental Health	1.15 (1.10, 1.20)	< 0.001
General Health	1.26 (1.21, 1.32)	< 0.001

*Comparison is gastrointestinal cancer vs. non-cancer, with non-cancer as the reference group.

803 **Poster Session**

Adequate racial and ethnic representation in cancer clinical trial enrollment: Real-world evidence from an NCI-designated cancer center in a minority-majority county.

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Background: Despite strides to increase diversity in clinical trial enrollment, significant racial and ethnic disparities persist. This limits the generalizability of clinical trial data and consequently the application of novel therapeutics for diverse populations. We hypothesized that enrollment of underrepresented minorities (URM) is feasible, with improved access to clinical care and a tailored clinical trial portfolio addressing the unmet needs in URM. **Methods:** A retrospective cohort study was conducted at the University of California Irvine Chao Comprehensive Cancer Center (CFCCC), an NCI-Designated cancer center located in Orange County (OC), CA (a minority-majority county). Clinical trial enrollment data from 2015-2023 through the CFCCC clinical research database was included. Collected data on patient demographics, tumor types, and trial enrollment was compared to data provided by the NIH SEER reports in both OC and United States. **Results:** Between 2015-2023, 2317 subjects were enrolled in clinical trials at CFCC. Demographics were as follows: White, Asian, Black, American Indian/Alaska Native, mixed/unknown race (66.4%/20.1%/2.4%/0.7%/0.7%/9.3%). Non-Hispanic vs Hispanic ethnicity (77.8%/20.5%). Female vs Male sex (47.6%/52.3%). Age <70 vs >70 years (73.3%/26.7%). Study Phase: Phase I/II (35.6%), Phase II (25.4%), Phase II/III (2.8%), Phase III (26.6%), and Phase IV (0.3%). Study sponsor: Industry (61.1%), Institutional (23.6%), National (14.8%), and Externally Peer Reviewed (0.3%). Most notably, the Asian enrollment (20.1%) and Hispanic enrollment (20.6%) exceeded demographic representation of Asians (17%, RR 1.18) and Hispanics (19%, RR 1.07) at CFCCC. Lastly, Asian patients enrolled at significantly higher rates in lung and liver cancer trials, while Hispanic patients and Black patients enrolled at higher rates in breast cancer and prostate cancer trials respectively. **Conclusions:** Our results demonstrate that robust URM clinical trial enrollment, particularly among Asian and Hispanic populations, is feasible at an NCI-designated cancer center despite minority-specific barriers. These findings suggest that minority populations can be optimally enrolled in clinical trials when these trials are accessible and tailored to the population at hand. Further research is necessary to investigate factors that influence clinical trial participation in the pursuit of equitable cancer care. Research Sponsor: None.

Comparative analysis of racial disparities in social determinants of health among patients with GI cancer in the All of Us dataset.

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Background: Social Determinants of Health (SDoH) are key factors in understanding disparities in cancer treatment and outcomes. Examining SDoH across racial and ethnic groups helps uncover the mechanisms behind disparities and identify solutions. This study compared SDoH factors between White (non-Hispanic) and Non-White (Asian, African American, Hispanic) patients diagnosed with Gastrointestinal (GI) cancers using a national dataset. **Methods:** We analyzed data from the All of Us Research Program. Adult patients with GI cancers, identified via ICD-10 and SNOMED codes, who had completed the SoDH surveys were included. Comparative analyses were performed between White and Non-White participants across SDoH constructs, including social/community construct, neighborhood environment, economic stability, and healthcare access using Chi-squared, T-test, and Mann-Whitney U tests as appropriate. **Results:** Of the 6,620 GI cancer patients identified, 1,831 completed the SDoH survey, of which 13.9% were Non-White. The most common cancers were colon and liver, with pancreatic, stomach, rectal, esophageal, and biliary tract cancers also included. White participants were older than Non-White participants (71.0 vs. 64.2 years, SD = 10.7 and 11.4) and reported higher levels of social cohesion (Mean (SD) = 3.92 (0.70) vs. 3.64 (0.76), $p < 0.001$) and social support (Mean (SD) = 3.92 (1.00) vs. 3.68 (1.11), $p < 0.001$). Non-White participants experienced greater perceived discrimination (Mean (SD) = 1.73 (0.80) vs. 1.54 (0.57), $p = 0.016$) and reported more neighborhood physical disorder ($p < 0.001$). Food insecurity (23.6% vs. 5.8%, $p < 0.001$) and housing quality issues (41.8% vs. 21.8%, $p < 0.001$) were more common among Non-White participants, who also faced more delayed medical care (45.9% vs. 32.5%, $p < 0.001$). **Conclusions:** This study highlights racial disparities in SDoH among GI cancer patients. Non-White participants experienced greater economic instability and neighborhood disorder compared to White participants. The All of Us dataset offers valuable data for examining SDoH in this population, though it may not fully represent the broader US population. As the dataset expands, it will provide more opportunities for comparative analyses among cancer patients across racial and demographic groups. Research Sponsor: None.

Construct	White Non-Hispanic (n = 1489)	Non-White (n = 255)	P-Value
Gender: Female, n (%)	827 (55.5)	167 (65.5)	p = 0.010
Social Disorder; mean (SD)	2.10 (0.22)	2.17 (0.28)	p = 0.002
Perceived Stress; mean (SD)	11.36 (7.12)	12.85 (7.99)	p = 0.003
Global Physical Health T-score (SD)	37.16 (1.89)	36.26 (2.08)	p < 0.001
Global Mental Health T-score (SD)	38.30 (1.76)	38.22 (1.86)	p = 0.520
Food Insecurity; n (%)	170 (5.8)	117 (23.6)	p < 0.001
Poor Housing Quality; n (%)	326 (21.8)	127 (41.8)	p < 0.001
Delayed Medical Care; n (%)	484 (32.5)	117 (45.9)	p < 0.001
Cannot afford specialist; n (%)	47 (4.2)	20 (11.9)	p < 0.001

805 **Poster Session**

Disparities in clinical trial enrollment among LGBTQ+ individuals and its impact on cancer treatment and management: A systematic review.

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Background: There is a significant gap in oncologic research, which arises from the exclusion of LGBTQ+ individuals from cancer clinical trials, compromising the development of personalized cancer therapies and limiting the generalizability of findings. Despite known health disparities in this population, including higher cancer risk factors and unique psychosocial stressors, data on clinical trial participation remains sparse. This study aims to elucidate the extent of underrepresentation of LGBTQ+ individuals in cancer clinical trials and examine its impact on treatment efficacy, safety, and overall management. **Methods:** We conducted a systematic review of literature and clinical trial databases, including PubMed, ClinicalTrials.gov, and other registries, focusing on cancer trials from 2010 to 2024. We extracted data on LGBTQ+ enrollment, analyzed demographic inclusivity criteria, and identified barriers to participation. Subgroup analyses evaluated the correlation between LGBTQ+ representation and treatment outcomes, considering variables such as trial design, recruitment strategies, and data collection on sexual orientation and gender identity. The review also examined policy changes and inclusivity initiatives aimed at improving trial access for this community. **Results:** The review identified a significant underrepresentation of LGBTQ+ individuals in cancer trials, with sexual orientation and gender identity data reported in only 6.2% of trials. Structural and sociocultural barriers, including heteronormative trial designs, lack of inclusive recruitment strategies, and pervasive discrimination, were frequently cited as impediments. Trials that included LGBTQ+ participants rarely conducted stratified analyses to assess differential treatment responses, leading to a paucity of data on drug safety, efficacy, and toxicity profiles specific to this population. The absence of these data may contribute to suboptimal therapeutic decision-making and potential adverse outcomes in clinical practice. **Conclusions:** The substantial underrepresentation of LGBTQ+ individuals in cancer clinical trials impairs the ability to develop inclusive, evidence-based oncology care. Addressing this gap requires systematic changes, including the integration of sexual and gender minority data collection, implementation of LGBTQ+-specific recruitment frameworks, and stratified analyses of treatment outcomes. Enhancing trial inclusivity will not only improve the external validity of oncologic research but also support the development of tailored therapeutic approaches that address the unique needs of LGBTQ+ cancer patients, ultimately advancing health equity in cancer care.
Research Sponsor: None.

Racial disparities in cancer stage at diagnosis among older adults with gastrointestinal cancers in the southeastern United States.

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Background: Prior studies have documented racial disparities in survival among older adults with various gastrointestinal (GI) cancers, but mechanisms remain poorly understood. We hypothesized that delayed diagnosis leading to advanced cancer stage at presentation could be a contributing factor to these disparities. **Methods:** We included adults ≥ 60 years with newly diagnosed GI cancers presenting for initial consultation at University of Alabama at Birmingham from October 2017 to April 2024. Using self-reported race and ethnicity, we stratified patients into Non-Hispanic Whites, Non-Hispanic Black, Hispanics and other category. For this analysis, we limited our cohort to the first two categories. Cancer stage was abstracted from the Electronic Health Record and followed American Joint Commission on Staging (AJCC) 7th manual. We performed two-sided Pearson's chi square test to test the association between stage at diagnosis and race. We performed this analysis for the overall cohort as well as for specific GI cancer types. The level of significance was set at $p < 0.05$. **Results:** A total of 1714 patients were included; Out of which 74% were non non-Hispanic white and 26% were African American. The median age at presentation was 68 years (interquartile range: 64 - 74), and 42% were females. The most common cancer type was colorectal (28%) followed by pancreatic (26%), hepatobiliary (22%), other GI cancers (14%) and gastroesophageal (10%). Overall, around 45% patients had Stage IV at diagnosis, followed by Stage III (26%), Stage II (17%) and Stage I (11%). We found that as compared to non-Hispanic Whites, non-Hispanic Blacks were more likely to have advanced stage disease, stage III (29% vs 25%) and stage IV (47% vs 45%), this association was statistically significant (Table). Furthermore, this association was most prominent for pancreatic cancers, where non-Hispanic Blacks were more likely to have advanced stage disease, stage III (21% vs 16%), and stage IV (50% vs 47%; $p < 0.05$). **Conclusions:** We report that among older adults with GI cancers, African Americans were more likely to have advanced stage at diagnosis compared to non-Hispanic whites, specifically in patients with pancreatic cancer. This finding may contribute to some of the racial disparities observed in survival outcomes in this population. Research Sponsor: None.

Contingency table showing frequencies (%) of cancer stages for non-Hispanic Whites and African Americans.

GI Cancer Stage	Non-Hispanic White	African American	p value
Stage I	138 (11%)	51 (11%)	<0.05
Stage II	236 (19%)	57 (13%)	
Stage III	321 (25%)	129 (29%)	
Stage IV	572 (45%)	210 (47%)	

807 **Poster Session**

Provider perspectives on non-medical barriers to GI cancer care at a regional cancer center.

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Background: Disparities in care for patients with gastrointestinal (GI) cancers are well-documented, but solutions remain elusive. Understanding the barriers faced by healthcare providers is essential to improving care equity. This study explored the perceptions of healthcare providers at a regional academic health system regarding barriers to care for GI cancer patients. **Methods:** A cross-sectional survey was conducted among healthcare providers, including physicians, advanced care providers (APPs), nurses, dietitians, social workers, care coordinators, and physical/occupational therapists who directly care for patients with GI malignancies. Survey items assessed specific barriers encountered in clinical practice and suggestions for improving care. Descriptive statistical analysis evaluated the quantitative data from the closed-ended questions across professional roles. A thematic analysis was performed on open-response items and assessed to further identify key barriers to care based on respondents' clinical roles and experiences. **Results:** Seventy-nine healthcare providers responded to the survey. Initial treatment planning was consistently identified as the phase with the most significant barriers across all disciplines. Health literacy was cited as the most critical barrier to diagnosis, while insurance coverage, financial constraints, geographic location, lack of transportation, and social support were the primary obstacles to treatment. The importance of these barriers varied by profession: physicians noted geographic and transportation limitations, while nurses and social workers emphasized gaps in health literacy, insurance, and social support. Systemic issues such as limited access to specialized care and healthcare personnel shortages were also noted. Notably, 55% of physicians spent 10-30% of their time addressing non-medical barriers to treatment, while 38% of nurses spent 30-50%. Care coordination across disciplines was commonly cited as a challenge, with many calling for improved communication and establishing multidisciplinary clinics. Food insecurity, housing stability, and language/cultural barriers were less prominent concerns at this institution. **Conclusions:** Oncology providers at a large regional cancer center spend significant time addressing non-medical barriers to treatment for GI cancer patients, particularly during initial treatment planning. The most common challenges reported were insurance, transportation, health literacy, and social support. We will use these findings to guide in-depth provider and patient interviews and develop a team-based, equity-focused intervention. These efforts aim to improve care delivery at our institution and serve as a model enhancing GI cancer care in broader healthcare settings. Research Sponsor: None.

Association between skeletal muscle morphometrics and overall survival among older adults with gastrointestinal cancers.

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Background: Older adults with gastrointestinal (GI) cancers face a higher risk of treatment toxicities and adverse survival outcomes. However, the variability in these outcomes remains poorly understood. Age-related changes, including the progressive loss of skeletal muscle and gain in adipose tissue, contribute to this heterogeneity. Recent studies highlight an association between skeletal muscle gauge (SMG) and overall survival (OS) in older cancer patients, but its relevance among those with GI cancers is unclear. **Methods:** We included adults aged ≥ 60 years with newly diagnosed GI cancers, enrolled in the University of Alabama at Birmingham (UAB) Cancer and Aging Resilience Evaluation (CARE) registry between October 2017 and April 2024. Skeletal muscle index (SMI; skeletal muscle area in m^2/height in cm) and skeletal muscle radiodensity (SMD; Hounsfield units) were derived from single-slice computed tomography (CT) staging scans at the third lumbar vertebra (L3) using validated methods. SMG was calculated as the product of SMI and SMD, reported in arbitrary units (AU). Participants were followed until death or the study cutoff date of July 10, 2024. Kaplan-Meier survival analysis was used to estimate OS by SMG status (low: < median; high: ≥ median). Cox proportional hazards models were employed to assess the relationship between SMG and OS, adjusting for age, sex, race/ethnicity, and cancer stage. **Results:** A total of 568 patients were included in the study, with a median age at diagnosis of 68 years (interquartile range, IQR: 64-74), 55% male, 73% non-Hispanic White, and 43% were diagnosed with Stage IV disease. Colorectal (33%) and pancreatic (27%) cancers were the most common diagnoses. The median SMG was 1,529 AU (IQR: 1,151–1,967). Over a median follow-up of 60 months, 367 patients (64%) died. Median OS was 19 months for those with low SMG and 26 months for those with high SMG (log-rank test, $p < .05$). After adjustment for confounders, SMG remained independently associated with OS (adjusted Hazard Ratio: 0.678 per IQR increase; 95% CI: 0.57– 0.807; $p < .001$), indicating that each IQR increase in SMG is associated with a decrease in all-cause mortality risk by 32.2%. **Conclusions:** Skeletal muscle gauge, an integrated measure of SMI and SMD, is significantly associated with improved survival among older adults with GI cancers. Further research is needed to determine whether SMG has a causal impact on survival in this population. Research Sponsor: None.

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Rate of gene fusion/rearrangement identification in GI cancers as sequencing evolved from limited panels to whole exome sequencing.

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Background: Gene fusion/rearrangements occur at a frequency of 0-15% in GI system cancers. Detection and treatment of fusion genes such as NTRK, FGFR2, NRG1 and RET fusions are included in NCCN guidelines and are commonly detected using next generation sequencing (NGS). NGS panels have continued to expand over time, which should result in increases in the identification of gene fusions. The aims of this study were to describe the percent of patients with advanced GI tumors undergoing NGS with identified gene rearrangements across time as panels expanded. **Methods:** The study included 7237 patients in the Kaiser Permanente Northern California Integrated Health System receiving care for advanced GI cancers from 2017 to Aug 2024, whose tumors samples were evaluated using NGS (Stomach=926, Colorectal=2792, Gallbladder=569, Esophagus=761, liver=218, Pancreas=1971). **Results:** A total of 231 patients (3.2%) were identified with a gene rearrangement, including 4.6% in stomach(N=43), 2.8% in colorectal(N=78), 8.9% in gallbladder(N=24), 5.3% in esophagus(N=40), and 1.9% in pancreatic (N=38*). Rearrangements included NTRK1/2/3, FGFR2/3, NRG, ALK, ROS, CLDN18-ARHGAP, RSPO2/3, DNAJB1-PRKACA, BRAF, MET and RET.* The numbers per year were (see table). **Conclusions:** The use of more comprehensive NGS panels such as WES did not appear to increase the detection rate for fusion/rearrangements. In resource limited settings, limited panels may suffice. The number of GI cancer patients getting NGS testing has increased over time. Research Sponsor: None.

Gene rearrangements/fusions identified per year.

YEAR	2018	2019	2020	2021	2022	2023	2024 (Jan – Aug)
Gene Rearrangement identified	29	37	43	55	54	62	38
TOTAL GI Cancer Samples Sequenced	610	858	900	1047	1249	1181	785
Percentage	4.75%	4.31%	4.78%	5.25%	4.32%	5.25%	4.84%

Management trends and outcomes of appendiceal cancers: A National Cancer Database study.

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Background: Appendiceal tumors encompass a variety of histological types that exhibit different behaviors. The current study aimed to investigate and compare overall survival, predictive factors for overall survival, and management of histological variations of appendiceal tumors. **Methods:** The National Cancer Data Bank (NCDB) database (2004–2020) was evaluated to include patients with appendiceal tumors histologies including goblet cell adenocarcinoma (GCA), neuroendocrine neoplasm (NEN), non-mucinous adenocarcinoma (NMA), and mucinous adenocarcinoma (MA). Univariate and multivariable analyses were conducted. **Results:** The GCA, MA, NEN, and NMA groups consist of 6,111, 16,471, 19,199, and 11,065 patients, respectively. The NMA and NEN groups had significantly lowest and higher overall survival compared to other types ($p<0.001$ for both) (GCC: 143.27 (140.23-146.30), MA: 111.91 (109.996-113.832), NEN: 170.88 (168.56-173.20), and NMA: 101.399 (99.132-103.666) months). The independent predicting factors of worse overall survival were age (HR: 1.03 (1.03-1.04)), black race (HR: 1.24 (1.08-1.43), ref: white race), fourth median income (HR: 0.84 (0.74-0.96), ref: first median), Charleson index comorbidity score (HR: 1.3 (1.22-1.38)), lymphovascular invasion (HR: 1.28 (1.15-1.42)), pathologic stage 2 (HR: 1.32 (1.12-1.57)), stage 3 (HR: 1.99 (1.66-2.40)), stage 4 (HR: 4.20 (3.47-5.07)), ref: stage 1, intraoperative systemic therapy (HR: 0.52 (0.39-0.70)), positive surgical margin (1.55 (1.39-1.73)), number of positive lymph nodes (HR: 1.06 (1.05-1.07)), moderately differentiated tumors (HR:1.43 (1.25-1.63)), poorly differentiated (HR: 1.94 (1.69-2.24)), undifferentiated (HR: 1.48 (1.14-1.91)), ref: well differentiated, and NMA type (HR:1.30 (1.10-1.55)). Furthermore, in terms of type of surgery, significantly higher patients of GCA (54.6%) and NMA (55.2%) groups underwent subtotal colectomy compared to other groups. Regarding systemic therapy, significantly higher patients of MA group received intraoperative, and both neoadjuvant and adjuvant therapy compared to other groups. Also, NEN group had significantly higher patients who underwent surgery with laparoscopic approach compared to others. **Conclusions:** This study reveals that different types of appendiceal tumors have distinct characteristics in regard to overall survival outcomes and treatment approaches. NENs showed the highest survival rates, while NMAs had the lowest. These results underscore the need for personalized treatment strategies for each tumor type to improve patient outcomes. Research Sponsor: None.

Treatment patterns in patients with advanced gastrointestinal stromal tumor in Japan: An administrative claims database study.

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Background: Gastrointestinal stromal tumors (GIST) are the most common mesenchymal tumors of the gastrointestinal tract. Tyrosine kinase inhibitors, including imatinib, sunitinib, and regorafenib, and the HSP90 inhibitor, pimitespib have been approved for advanced GIST in Japan, but few studies have examined the nationwide trend in treatment patterns. This study aimed to evaluate the real-world treatment patterns for advanced GIST in Japan. **Methods:** This retrospective cohort study used the Japanese hospital-based administrative claims database provided by Medical Data Vision Co., Ltd. (Tokyo, Japan). Patients diagnosed with advanced GIST according to ICD-10 codes who received any approved standard drug in Japan, such as imatinib, sunitinib, regorafenib, or pimitespib (Approved in June 2022) between July 2013 and March 2023 were included. The primary endpoint was the proportion of patients with GIST who were treated with each approved standard drug. Secondary endpoints included treatment patterns, time to treatment failure (TTF), overall survival, adverse events, and the proportion of patients who underwent surgical intervention for GIST while taking standard drugs. **Results:** This study included 2702 patients with advanced GIST. The median age was 70.0 years, 17% of the patients were ≥80 years, and 59% were male. The most common primary tumor sites were the stomach (37%), small intestine (25%) and large intestine (12%). The proportions of patients with GIST treated with imatinib, sunitinib, regorafenib, and pimitespib were 94% (2536 patients), 26% (702 patients), 14% (376 patients) and 1% (37 patients), respectively. The most common first-line regimen was imatinib (2485/2702 patients; 92%), second-line regimen was sunitinib (560/672 patients; 83%), third-line regimen was regorafenib (200/317 patients; 63%), and fourth-line regimen was imatinib (55/119 patients; 46%). The median TTF was 30.4 months (95% CI 27.4–32.9) with imatinib, 6.4 months (95% CI 5.6–7.7) with sunitinib, 5.2 months (95% CI 4.4–6.2) with regorafenib and 2.5 months (95% CI 1.7–3.1) with pimitespib. The proportions of patients who underwent surgical intervention for GIST was 15% (403/2702 patients), and curative surgery was 10% (273/2702 patients). **Conclusions:** This is the first real-world study to provide insights into the patient characteristics and current treatment patterns for advanced GIST in Japan. The treatment patterns are consistent with the Japanese Clinical Practice Guidelines. Research Sponsor: Taiho Pharmaceutical.

Current status and treatment outcomes of intraductal papillary mucinous carcinoma in Korea: A multi-center cohort study.

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Background: The spectrum of intraductal papillary mucinous neoplasm (IPMN) varies from benign cysts to intraductal papillary mucinous carcinoma (IPMC). There is a lack of large-scaled data on IPMC and high-grade dysplasia. We aimed to study the real-world status and treatment outcomes of IPMC in South Korea. **Methods:** This is a retrospective, multicenter study conducted in South Korea. We reviewed the electronic medical records of patients who were histologically diagnosed with IPMC or high-grade (HG) dysplasia between 2011 and 2020, and analyzed the baseline characteristics, treatment patterns, and survival. **Results:** A total of 388 patients were eligible from six institutes (145 female [37%]; median age 67.8; 259 IPMC [67%] vs 129 HG dysplasia [33%]). The median overall survival for IPMC patients was 75 months, compared to 133 months for HG dysplasia. In IPMC group, 217 (84%) were underwent surgical resection and 42 (16%) did not underwent surgery due to general condition or advanced stage whose median overall survival was 83 months and 49 months respectively. In resected IPMC, 104 (48%) received adjuvant chemotherapy (gemcitabine single 48%, 5FU+leucovorin 33%, others 19%) and 66 (30%) experienced recurrence and median recurrence-free survival (RFS) was 20 months. Among IPMC group, 11 (4.2%) was in advanced stage and eight of those received palliative chemotherapy (FOLFIRINOX 62.5%; gemcitabine single 25%; gemcitabine with erlotinib 12.5%). In HG dysplasia, 108 (84%) patients underwent surgery, and 21 (16%) patients did not undergo surgery whose OS were 171 and 133 months, retrospectively. In patients with resected HG dysplasia, 29 (27%) experienced recurrence and median RFS was 21 months. **Conclusions:** IPMC generally showed longer survival and a lower recurrence rate than those of historical data in PDAC, with HG dysplasia demonstrating even better outcomes. In real world, similar chemotherapy regimens used for PDAC were applied to IPMN patients. Research Sponsor: None.

Treatment status of intraductal papillary mucinous carcinoma and high-grade dysplasia.

	IPMC(259)	IPMN-HG(129)	Total(388)
Median overall survival (month)	75	133	85
Treatment OP	217(84%)	108(84%)	325(84%)
Adjuvant chemotherapy	104(48%)	19(18%)	123(38%)
Recurrence rate	66(30%)	29(27%)	95(29%)
Median recurrence-free survival (month)	20	19	21
Palliative chemotherapy	8(3%)	NA	
Progression rate	5(63%)	NA	
Progression-free survival (month)	13	NA	

813 **Poster Session**

Patient-reported outcomes in GI cancer: Leading change to meet the needs of the gastrointestinal cancer community.

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Background: According to the World Health Organization International Agency for Research on Cancer, globally gastrointestinal (GI) cancers account for 1 in 4 cancer cases and 1 in 3 cancer deaths, with an estimated 4.8 million new cases and 3.4 million GI cancer deaths reported annually. Understanding the global burden of GI cancers and the importance of understanding the real-world impact of a GI cancer diagnosis, the GI Cancers Alliance (GICA) conducted a 12-month patient-reported outcomes (PRO) research initiative that included a workshop at the 2023 American Society of Clinical Oncology (ASCO) annual meeting. In addition, our PRO research was published in the Journal of Clinical Oncology (J Clin Oncol 42, 2024 suppl 3; abstr 731) with a poster presented at the 2024 ASCO Gastrointestinal Cancers Symposium. **Methods:** Based on our 2022-2023 PRO research of 1,122 patients highlighting the unmet needs and critical gaps in services of the GI cancers community, we developed a series of follow-up initiatives. Nutrition education 17%, biomarker/precision medicine education 19%, patient-centered care education 11% and caregiver support 16% were four of the highest levels of unmet need based on our global research. Subsequently, in 2023 we created the 'GI Cancers Patient & Caregiver Advisory Board' representing all GI cancers. The board hosts educational forums supporting patients and caregivers throughout the care continuum. Our pilot nutrition program launched at the 2024 ASCO annual meeting, consisting of in-person and online video components, and an extensive nutrition guide with recipes tailored for GI cancer patients. In 2024, we partnered with our gastric cancer advocates to develop 'Test Your Biomarkers' enhancing education and awareness of biomarker testing and precision medicine. **Results:** Implementing direct patient feedback from our published research, we are creating actionable solutions to unmet needs and gaps in services. Our 'Patient & Caregiver Advisory Board' continues to amplify the patient and caregiver voice participating in global engagement opportunities. GICA provided 10,000 nutrition guides to community oncology centers, and our online messaging reached 250,000 individuals across digital platforms and partner websites in the first three months from launch. In the first six months of the 'Test Your Biomarkers' campaign our messaging reached 1.7 million individuals across all digital platforms, campaign, and partner websites. **Conclusions:** Our PRO research provides the roadmap for new initiatives and programming to meet the needs of our GI cancer community. The GI Cancers Alliance will proceed to collaborate with our 100+ partner organizations to activate change by targeting the needs of our patient community while eliminating critical gaps in services. By doing so, we will continue to provide greater patient impact through the amplification of the patient voice. Research Sponsor: None.

Impact of next-generation sequencing in advanced gastrointestinal cancer.

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Background: Next-generation sequencing (NGS) is a cutting-edge technology that allows for rapid DNA and RNA sequencing, identifying key mutations in cancer patients. It provides both diagnostic and predictive data for treatment decisions. Approximately 75.6% of US oncologists use NGS testing for patients with advanced solid tumors.This study retrospectively evaluates the impact of NGS testing on treatment outcomes for advanced GI cancer patients at a community-based cancer center in Florida. **Methods:** Between March 2019 and September 2023, 267 patients with stage 3 or 4 solid malignancies underwent NGS testing. Of these, 77 patients with advanced GI cancer were included in the study. Nearly half (48%) of the patients were diagnosed with advanced-stage colon cancer.NGS testing was performed using tissue samples (79.22%), blood samples (16.88%), or both (3.90%). Demographic and clinical outcomes, including time to diagnosis, treatment initiation, and survival, were collected and analyzed. **Results:** The median age of the study population was 70, with 88.31% of patients insured through Medicare. The cohort was composed of 61.04% males. Targetable mutations were identified in 59.74% of patients, leading to FDA-approved targeted therapies for 44.16%, and clinical trial eligibility for 79.22%. 52 patients were alive at the time of this analysis; those with biomarkers identified through NGS testing had a median treatment duration of 11 months regardless of whether testing impacted choice of therapy; whereas, patients without identified biomarkers had a median duration of 7.5 months. NGS testing impacted first-line therapy in only 6.49% of all patients; however, 80% of those patients were alive at the time of this analysis with a median duration of response of 5 months. **Conclusions:** NGS testing has become a cornerstone of oncologic care for advanced solid tumors; however, in this cohort of advanced GI malignancies, NGS results influenced first-line treatment decisions in only 6.49% of cases. Although the study included a small sample size, it reflects a real-world community setting where access to therapies and insurance coverage varies. For example, though patients may have targetable novel therapies available for their disease, the indications for such therapies may not be in the first line setting or covered by insurance. Advancements in NGS targeted therapies has greatly impacted the care of other solid malignancies such as non-small cell lung cancer; however, this study highlights that in the field of GI oncology, there is a large clinically unmet need of identifying newer biomarkers that are clinically relevant and impactful in earlier lines of oncologic care. Research Sponsor: None.

815 **Poster Session**

The efficacy of anamorelin in gastrointestinal cancer with cancer cachexia: Single center experience.

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Background: Cancer cachexia is a catabolic syndrome that causes weight loss and loss of appetite in cancer patients. Anamorelin (ANAM) is an oral drug approved in Japan to treat cancer cachexia by selectively activating ghrelin receptors. There is limited real-world data available on the efficacy of ANAM for patients with gastrointestinal cancer. We assessed its efficacy and identified associated clinical features. **Methods:** We retrospectively collected data from medical records and questionnaires on gastric and colorectal cancer patients who received ANAM between April 2021 and March 2023. We evaluated the change in body weight, appetite, and oral intake from the time of initiation of ANAM. We judged the responder who gained more than 0.5 kg in body weight, improved appetite, and increased oral intake, respectively. The association between clinical features and the responders was also assessed using multivariate logistic regression analysis. **Results:** 72 patients were included in this study. Characteristics of the patients: median age: 68 years old (range 38-85), male/female: 35/37, gastric/colorectal cancer: 39/33, PS 0-1/2 or more: 63/9, median body mass index (BMI): 18.0 kg/m^2 (range 12.3-27.9), number of prior chemotherapy regimens 1/2/3 or more: 17/22/33, ascites none/mild/moderate/severe: 42/19/7/4, median serum albumin levels (Alb): 3.2 g/dl (range 2.0-4.2). The median treatment duration of ANAM was 47.5 days (3-450 days). 57 patients (79.2%) continued ANAM for more than 3 weeks, and 9 patients for more than 12 weeks. The main reasons for discontinuation were progressive disease (39.7%), ineffective (12.7%), effective (11.1%) and adverse events of ANAM (11.1%). 33 patients (45.8%) gained weight, 41 patients (56.9%) improved appetite and 34 patients (47.2%) increased oral intake. There was a correlation found between BMI $<$ 18.0 kg/m^2 and the responders (OR 3.43, $p = 0.041$). **Conclusions:** In this study, 68.0% of patients who received ANAM for showed response, that is, gained body weight, or improved appetite, or increased oral intake. And these observations suggested that BMI was associated with effects of ANAM. Further studies involving more patients needed. Research Sponsor: None.

Utilizing an automated, high-throughput platform to generate comprehensive clinical & molecular datasets to enable a real-world evidence approach for improving care of patients with gastrointestinal malignancies.

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Background: More than 44,000 patients are seen at MD Anderson Cancer Center annually. However, using the diverse, mostly unstructured, data from these patient encounters has been challenging and required manual chart reviews. In 2018, MD Anderson and Palantir Technologies (Denver, CO) began developing a unified, cloud-based, graphical user interface clinical informatics platform to extract, structure, and integrate data from the different data sources that comprise the electronic health record (EHR). Here, we describe our experience using this novel platform to apply a real-world evidence (RWE) approach to study patients with gastrointestinal (GI) malignancies. **Methods:** Institutional Review Board approval for retrospective chart review of patients with GI malignancies was previously obtained. The Foundry platform was used to incorporate more than 150 datasets, including structured data elements like lab values, unstructured data as the full text of clinical notes, and natural language processing (NLP) derived datasets. The datasets include unique patient identifiers to allow the merging of demographic, clinical, molecular, and outcomes information. The platform allows processing of the note text through NLP to extract non-discrete data elements into a discrete form. In addition, it continuously updates new data on daily bases, allowing the inclusion of new patients' information in an automated fashion. **Results:** From 2,013,048 patients with date of diagnosis ranging from 1944 to 2024, we have created datasets for colorectal adenocarcinoma (CRC, >50,000 patients, >8,000 with molecular data), pancreatic adenocarcinoma (PDAC, >13,000 patients), and appendiceal adenocarcinoma (AA, >3,000 patients). More than 50 variables have been integrated, including demographic information, stage, grade, overall survival, and molecular information. Focused manual validation of the automated extraction across the cohorts consistently demonstrated an accuracy of over 94%. Work is underway to extract additional features including DFS and PFS, sites of metastasis, and to build out additional cohorts for biliary tract, upper GI, and neuroendocrine tumors. Initial discovery efforts have already led to multiple publications including discovery of molecular causes of racial and ethnic disparities in CRC, survival impact of *KRAS* and co-mutations in PDAC, and prognostic utility of serum tumor markers in AA. **Conclusions:** Utilizing an automated, highly dynamic platform allowed integration of comprehensive datasets for multiparameter oncology data in patients with GI malignancies. This resulted in dramatic acceleration of cohort identification, outcomes analysis, and enabled utilizing a data-driven approach to guide decision making in an effort to enhance and optimize outcomes. Research Sponsor: national cancer institute; K22 CA234406; the Cancer Prevention & Research Institute of Texas; RR180035; the Cancer Prevention & Research Institute of Texas; RP240392; Conquer Cancer, the ASCO Foundation; CDA-7604125121.

817 **Poster Session**

The impact of prediabetes on cerebrovascular risk among patients with gastrointestinal malignancies.

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Background: Cancer screening ingastrointestinal (GI) malignancies has translated into an improvement in overall survival due to detection at an earlier stage. The majority of survivors (up to 67%) are over the age of 65. The impact of metabolic profiles on cerebrovascular events amongst this population of cancer survivors is unclear. **Methods:** Data from the NIS database from 2016 to 2020 were reviewed for patients with seven GI malignancies. Analysis was performed based on the presence or absence of prediabetes using ICD-10 codes. Baseline characteristics, comorbidities, and cerebrovascular outcomes were studied in these 2 cohorts. **Results:** 1,532,250 hospitalizations of patients with seven GI malignancies, namely colorectal cancer (CRC), pancreatic cancer, hepatocellular carcinoma (HCC), gastric cancer, small intestinal cancer, cholangiocarcinoma, and esophageal cancer, were evaluated and divided into two subgroups: prediabetics (15,610) and non-prediabetics (1,516,640). Prediabetics with GI malignancies as compared to non-prediabetics with GI malignancies were noted to be older (67.7 years vs. 65.64 years), with higher total hospitalization charges (USD 87,267 vs. USD 80,173), greater rates of elective admissions (34.92% vs. 25.89%) but a lower length of stay (6.018 days vs. 6.53 days). Prediabetic patients with GI malignancies were noted to have a higher prevalence of obesity (18.64% vs. 7.69%), dyslipidemia (52.99% vs. 26.16%), and hypertension (49.59% vs. 39.37%). Prediabetics with CRC, as compared to non-prediabetics, had higher adjusted odds ratios (aORs) of occlusion/stenosis of precerebral or cerebral arteries not leading to cerebral infarction, and cerebral infarction, with aORs of 2.03 (95% CI 1.24-3.30, p 0.004), and 1.63 (95% CI 1.01-2.63, p 0.045) respectively. Additionally, prediabetics with pancreatic cancer had a higher aOR of cerebral infarction as compared to non-prediabetics, with aOR of 2.42 (95% CI 1.70-3.44, $p<0.001$). **Conclusions:** Prediabetic patients with CRC had higher aORs of cerebral infarction and occlusion/stenosis of precerebral arteries not leading up to cerebral infarction, and prediabetics with pancreatic cancer had twice the risk of cerebral infarction as compared to non-prediabetics. This study highlights the impact of prediabetes, a modifiable risk factor, on cerebrovascular outcomes in patients with GI malignancies. Research Sponsor: None.

Surgical treatment in patients with advanced gastrointestinal stromal tumours (GISTs) in second-line and later stages: A single-centre cohort study.

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Background: Gastrointestinal stromal tumour (GIST) after imatinib treatment failure usually progresses rapidly with limited overall survival. Previous studies have suggested that surgery is beneficial for advanced GISTs in disease control to imatinib, but the role of surgery in > = 2 line patients remains unclear. **Methods:** Imatinib-resistant patients with advanced GIST between May 2009 and Nov. 2023 were included. Surgical treatments included tumour reduction, radiofrequency ablation and TACE. The primary endpoint was overall survival (OS) and secondary endpoints were time to second-line treatment failure (TTF) and postoperative morbidity. To balance clinicopathological characteristics, 1:1 propensity score matching (PSM) was used. **Results:** A total of 242 patients with advanced GIST treated with 2nd-line or later drugs were included, including 125 in the surgical group and 117 in the non-surgical group. The median follow-up was 30.3 months. OS was significantly better in the surgical group than in the non-surgical group (pre-PSM: 51.7m vs 35.9m , p = 0.001; post-PSM: 50.8m vs 38.8m, p = 0.043). Univariate analysis showed that surgical treatment (HR 0.487, 95% CI 0.320-0.740, p = 0.001), age ≤60 years (HR 0.640, 95% CI 0.411-0.997, p = 0.049) and low tumour burden (HR 0.491, 95% CI 0.293-0.822, p = 0.007) were associated with prolonged OS. Multivariate analysis indicated that surgical treatment (HR 0.541, 95% CI 0.348-0.843, p = 0.007) and low tumour burden (HR 0.510, 95% CI 0.300-0.865, p = 0.012) were significantly associated with OS benefit. Patients (n = 84) who had surgical treatment during second-line treatment had longer TTF compared to patients (n = 158) who had TKI only (14.2m vs 8.1m , p = 0.003). There were 29 (14.8%) postoperative complications with Clavien-Dindo classification ≥3, including 5 deaths. **Conclusions:** Surgical treatment can provide a survival benefit for patients with second-line and later advanced GIST, but further studies are needed for a more detailed indication of the surgical population. Research Sponsor: None.

819 **Poster Session**

Risk of second primary cancer among patients with a first primary appendiceal adenocarcinoma.

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Background: Incidence rates of rare appendiceal cancers are increasing across the United States. Although this is suggestive of a growing population of appendiceal cancer survivors over time, the risk of second primary cancers specific to patients with a first primary appendiceal adenocarcinoma remains unknown. **Methods:** We conducted a population-based study of adults (age ≥18 years) diagnosed with a first primary appendiceal adenocarcinoma from 1992 to 2021, using data from the Surveillance, Epidemiology, and End Results (SEER) Program. We estimated cumulative incidence of second primary cancer, defined as any primary cancer diagnosed at least 3 months after appendiceal adenocarcinoma, with Fine and Gray methods to account for the competing risk of death. For comparison to the general population, we also estimated standardized incidence ratios [SIRs], or the observed number of second cancers relative to the expected number based on age-, sex-, and race-specific incidence rates. **Results:** A total of 5,827 adults diagnosed with a first primary appendiceal adenocarcinoma (mean age 58.0 years; 51.8% female; 68.5% non-Hispanic White) were identified during the study period, of whom 418 developed a second primary cancer. The most common types of second primary cancer were prostate (17.7%), colorectal (14.4%), female breast (10.5%), lung (8.4%), and melanoma (6.0%). At 10 and 20 years after appendiceal adenocarcinoma diagnosis, the cumulative incidence of second primary cancer was 7.5% (95%CI 6.8%-8.4%) and 11.7% (95%CI 10.5%-12.9%), respectively. Cumulative incidence significantly differed by sex ($p<0.01$) and age at appendiceal adenocarcinoma diagnosis (early-onset [age<50] *vs* late-onset; $p<0.01$). For example, at 10 years post-diagnosis, the cumulative incidence of second primary cancer was 6.4% (95%CI 5.4%-7.4%) for females and 8.8% (95%CI 7.6%-10.1%) for males. The overall SIR for any second primary cancer was 1.12 (95%CI, 1.02-1.23)—corresponding to 14.8 excess cancers per 10,000 person-years. SIRs for common types of second primary cancers were: 1.07 (95%CI 0.84-1.35) for prostate, 2.12 (95%CI 1.63-2.70) for colorectal, 0.91 (95%CI 0.68-1.19) for female breast, 0.81 (95%CI 0.58-1.09) for lung, and 1.20 (95%CI 0.79-1.73) for melanoma. **Conclusions:** Approximately one in every 13 adults diagnosed with a first primary appendiceal adenocarcinoma in the United States will be diagnosed with a second primary cancer within 10 years. After diagnosis of an appendiceal adenocarcinoma, patients have a 12% excess incidence in second primary cancers and double the excess incidence in second primary colorectal cancers compared to the general population. Implementation of cancer prevention and early detection strategies (e.g., colonoscopy screening, genetic testing) for patients with an appendiceal adenocarcinoma is a clinical priority. Research Sponsor: Appendix Cancer Pseudomyxoma Peritonei (ACPMP) Research Foundation; National Cancer Institute/U.S. National Institutes of Health; P50 CA236733; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

820 **Poster Session**

Evaluating the clinical benefits of anti-cancer drugs approved by Food and Drug Administration for gastrointestinal neoplasms for the past two decades.

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Background: The number of FDA-approved anticancer drugs have increased in recent years with a more use of surrogate endpoints. However, it is unclear whether the same trend applies to GI cancers. Therefore, we aimed to evaluate the trend in FDA-approved GI cancer drugs over the past two decades and the associated clinical benefit. **Methods:** The FDA's online database was searched for drugs approved for GI cancers from 2006 until 2024. Primary and secondary endpoints along with trial characteristics were extracted. ESMO Magnitude of Clinical Benefit Scale (ESMO-MCBS) was assessed by two blinded investigators with a third resolving disagreements. We divided the study into period A (01/2006-12/2014) and period B (01/2015-08/2024). **Results:** Since 2006, FDA approved 62 regimens for GI cancer based on 70 trials, mostly (n= 45;72.6%) in period B. Most trials (n=44) were phase III with only 26 phase II: 22 single arm, and 4 randomized. All the 22 single-arm phase II trials were approved in period B with two randomized phase II trials approved in each period. Approvals based on phase III trials dropped by 33.5% (from n=15 to n=29) between the two periods. Half (n=34) of the approvals were based on overall survival (OS) benefit, while the use of objective response rate (ORR) as an endpoint has increased from 11.8% (n=2) in period A to 45.3% (n=24) in period B. Quality of life (QoL) was assessed only in 27% of the studies and found benefit only in 22.7%. The mean gain in OS was similar across both periods; 2.3 months [1.2-4.7]. Out of the 19 accelerated approvals, 17 were in period B, of which 23% (4/17) were based on secondary endpoint, 18% (3/17) were withdrawn, and 59% (10/17) received full approval. Substantial clinical benefit (ESMO score 4-5) was seen in 11.8% of the trials in period A and 19.6% in period B, with 17.4% overall from 2006-2024. **Conclusions:** Despite rising FDA approvals for GI cancer drugs since 2006, only 17.4% demonstrated substantial clinical benefit based on ESMO-MCBS. Recent approvals rely more on single-arm phase II studies and ORR, often without OS or QoL benefit. These trends underscore the need for more rigorous clinical trials that prioritize meaningful endpoint to ensure that new therapies provide real clinical value to the patient. Research Sponsor: None.

Frailty assessed by a 10-item index from comprehensive geriatric assessment (FI-CGA-10) versus oncologist-assessed performance status (PS) in older adults with gastrointestinal (GI) cancer.

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Background: A FI-CGA-10 is a recently developed measure of frailty in the geriatric oncology setting [Oncologist, 26, e1751 (2021)]. Our objective was to compare the multidimensional frailty assessment by FI-CGA-10 with PS assessed by a primary oncologist in older adults with cancer. **Methods:** This study included 790 older adults with GI cancer who underwent a CGA before cancer treatment decisions at a geriatric oncology service between September 2018 and May 2024, and whose PS was documented in the electronic medical record by a primary oncologist. Fitness and frailty level were evaluated using the FI-CGA-10, which assesses 10 domains: cognition, mood, communication, mobility, balance, nutrition, basic and instrumental activities of daily living, social support, and comorbidity. Deficits in each domain were scored as 0 (no problem), 0.5 (minor problem), and 1.0 (major problem). FI-CGA-10 scores (range 0-1) were calculated by dividing the sum of the scores for each domain by 10 and then categorized as fit (<0.2), pre-frail (0.2–0.35), and frail (>0.35). The strength of the ordinal association between the frailty category and each CGA domain score was assessed using Spearman's non-parametric correlation coefficient (rho). **Results:** The median age was 79 years; 60% were male, 62% had GI tract cancer (esophageal, gastric, colorectal), 20% had hepatobiliary cancer, 15% had pancreatic cancer, and 46% had stage 4 disease. Overall (n=790), 25% of patients were fit, 40% were pre-frail, and 35% were frail (Table). Among patients with PS 0-1 (n=633), 30% were classified as fit, 48% as pre-frail, and 22% as frail. Among PS 0-1 patients, Spearman's rho was greater than 0.5 for the association between the frailty category (fit, pre-frail and frail) and cognition, IADL, and mobility domains. In this subcohort (PS 0-1), the proportion of patients with a cognition domain score of 0.5 (mild cognitive impairment) or 1.0 (dementia) was 7%, 30%, and 78% in the fit, pre-frail, and frail groups, respectively; the proportion of those with an IADL domain score of 0.5 (OARS IADL = 12-13) or 1.0 (OARS IADL ≤ 11) was 5%, 46%, and 92%, respectively; the proportion of those with a mobility domain score of 0.5 (0.8 m/s ≤ gait speed < 1.0 m/s) or 1.0 (gait speed < 0.8 m/s) was 45%, 85%, and 98%, respectively. **Conclusions:** In this study, 70% of older adults with GI cancer whose PS was scored as 0-1 by a primary oncologist were classified as pre-frail or frail when assessed using FI-CGA-10. Among patients with PS 0-1, there were strong correlations between the frailty levels and impairment levels in the cognition, IADL, and mobility domains, indicating that these domains are major contributors to the frailty classification (fit, pre-frail, and frail).
Research Sponsor: None.

All patients (n=790)	Fit	Pre-frail	Frail
PS 0 (n=351), n (%)	145 (41)	163 (46)	43 (12)
PS 1 (n=282), n (%)	45 (16)	141 (50)	96 (34)
PS 2-3 (n=157), n (%)	4 (3)	18 (11)	135 (86)

A qualitative study of the home-based time burdens faced by persons affected by GI cancer.

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Background: As cancer care is increasingly delivered at home, more tasks and responsibilities fall on patients and informal care partners. This is especially relevant for people with GI cancer who often have specific home-based care needs such as feeding, ostomy care, 5-fluorouracil chemotherapy, and complex medication management. Home-based time costs are under-counted in current measures of time toxicity that only include care received in formal health-care settings. **Methods:** We conducted semi-structured interviews with patients with GI cancer and their care partners at a single tertiary cancer center in MN, USA from March-Oct 2023. Interviews explored cancer care tasks at home, associated time burdens, and how they compare to in-facility care. We analyzed interview transcripts using a grounded theory approach to identify themes. **Results:** We included 33 individuals (11 aged >60y, 21 female, 25 white) including 15 patients (8 colorectal cancer, 8 ECOG 0) and 18 care partners (15 lived with the patient, 7 working at least part-time). We identified 5 themes (Table). **Conclusions:** This foundational work characterizes time burdens of home-based cancer care for persons affected by GI cancer. In addition to providing the first input on incorporating home-based time burdens into objective measures of time toxicity, we highlight the complexity of assessing time burden which is context and role-dependent. Research Sponsor: Minnesota Cancer Clinical Trials Network; American Cancer Society; Institutional Research Grant #129819-IRG-21-049-61-IRG138; Pancreatic Cancer Action Network; Minnesota Colorectal Cancer Research Foundation.

Theme	Subtheme	Quote
Unexpected home-based care is time burdensome	Challenge with home as a site of healthcare, difficult transition from hospital to home	"What they would do at the clinic behind the scenes she was doing on my kitchen table." (Patient)
Other burdens compound time burdens	Logistic/administrative, emotional, symptom	"The time spent on administrative things and coordinating [...] are more stressful than going to the clinic." (Patient)
Time burdens evolve over the disease course and differentially impact patients and care partners	Cancer care constantly changes, discomfort with some tasks at home, less support provided if more competent, home-based care allows for connection	"They're in a season of remission and [...] we want to live as normal a life as we can. And then there are other seasons where it's the absolute opposite." (Care partner)
Factors influencing the choice of home-based care	Need to manage family duties, supportive care partners matter, value connection with nurses, decreased travel	"There is a different cost when you start doing things at home, especially when you have small children because if they're witnessing it, that's emotional." (Patient)
Home-based care is generally perceived as less time-burdensome than similar cares provided in-facility	Different opinions on whether home-based care should be included in time toxicity measure	"If you're just talking about video visits I wouldn't consider that time toxic. But if I'm going [to the clinic], I've pretty much lost the whole day." (Patient)

823 **Poster Session**

A systematic review and meta-analysis on evaluating the efficacy and safety of netupitant/palonosetron compared to aprepitant for the prevention of chemotherapy-induced nausea and vomiting in patients undergoing emetogenic chemotherapy.

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Background: NEPA stands as the exclusive oral fixed combination antiemetic agent, comprising the highly selective NK1 receptor antagonist (RA) netupitant (300 mg) and the clinically distinct 5-HT3 RA, palonosetron (0.5 mg). Administered in a single dose, NEPA concurrently modulates two critical antiemetic pathways, offering a more convenient and straightforward alternative for sustained protection against chemotherapy-induced nausea andvomiting (CINV). Numerous studies have assessed NEPA's efficacy relative to other Neurokinin (NK) 1 receptor antagonists (RAs), notably aprepitant (APR), widely utilized for CINV. Our study critically examines the efficacy and safety of NEPA versus APR for preventing CINV in patients receiving emetogenic chemotherapy, drawing insights from recent studies. **Methods:** Studies comparing NEPA versus APR for the prevention of CINV in patients receiving emetogenic chemotherapy until August 2024 were systematically searched on Medline, Embase, and Cochrane Library. Notably, no systematic review or meta-analysis of pooled data from existing studies is currently present in the literature. **Results:** Following screening, four studies were deemed eligible for inclusion in the meta-analysis, encompassing a total of 2079 patients. The NEPA group comprised 1047 individuals, while the APR group comprised 1032 subjects. The primary outcome included the comparison of subjects experiencing no nausea in overall phases. Secondary outcomes involved subjects with no emesis and no nausea in the acute and delayed phases. A significant reduction in the risk of overall nausea in the NEPA group was observed compared to the APR group after Moderately Emetogenic Therapy (MEC) or Highly Emetogenic Therapy (HEC) [Risk Ratio (RR) of 1.07, 95% Confidence Interval (CI) = 1.01-1.13, P = 0.01). A trend of reduced emesis during both acute [RR = 1.08, 95% CI = 0.95-1.22, P = 0.24] and delayed periods [RR = 1.03, 95% CI = 0.99-1.07, P = 0.15] was observed in the NEPA group, but the results were not statistically significant. Significant nausea in the acute and delayed post-chemotherapy periods exhibited a similar trend (RR = 1.02, 95% CI = 0.98-1.06, P = 0.43) and (RR = 1.04, 95% CI = 0.98-1.10, P = 0.23) respectively, with non-significant statistical outcomes. **Conclusions:** Our analysis suggests that the use of NEPA results in better control of nausea in overall phases compared to Aprepitant for the prevention of CINV in patients receiving emetogenic chemotherapy. However, NEPA demonstrates comparable efficacy to aprepitant in achieving control of nausea and emesis when the symptoms are stratified in acute and delayed phases. Nevertheless, further studies are warranted for a comprehensive comparison between NEPA and Aprepitant for the prevention of CINV. Research Sponsor: None.

The presenting symptoms of patients with appendiceal malignant tumors.

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Background: Timely detection of appendiceal malignant tumors is a clinical priority because of the propensity for this malignancy to metastasize to the peritoneal cavity. Up to one in every 2 patients with this rare tumor type will present with distant metastatic disease, owing to there being no standardized screening tests for early detection of primary appendiceal cancers. Case reports have pointed to broad/non-specific symptoms of patients diagnosed with appendiceal malignant tumors, yet information is needed from large cohorts to support prompt/accurate clinical diagnoses. **Methods:** We analyzed data from patients with a pathologically-confirmed first primary appendiceal malignant tumor (AC) who prospectively enrolled in the nation-wide Genetics of Appendix Cancer [GAP] clinical cohort study (Clinicaltrials.gov, NCT05734430) between November 2022 and May 2024. The primary outcome was presenting symptoms that led to AC diagnosis. Symptom effects and interactions with diagnosis age/year, sex, body mass index (BMI), and tumor histology, were quantified with negative binomial regression models and presented as incidence rate ratios (IRRs) and 95% confidence intervals. **Results:** Of the 352 patients included in our analyses (median [IQR] age, 51.0 [42-59] yr), 77.6% were female and 96.6% had an adenocarcinoma histology. Seventy-seven percent of patients (n=270) reported one or more presenting symptoms, of whom 55.2% (n=149) experienced these symptoms for 3+ months prior to diagnosis. On average, patients reported 3 symptoms (mean: 2.9, SD 3.0). The most prevalent symptoms among males and females were abdominal pain (57.1%), bloating/distension (32.1%), pelvic pain (18.5%), and abdominal/pelvic mass (18.2%). Overall, patients with early-onset AC [age<50] more commonly presented with symptoms versus cases with late-onset AC (82.9% *vs* 71.7%, p=0.01). This finding persisted in adjusted models—the number of symptoms was highest among younger patients, rapidly decreased and plateaued around age 50 (IRR 0.97, 95%CI 0.95-0.99, p=0.002), and slowly increased again with older age (IRR 1.03, 95%CI 1.00-1.05, p=0.037). Patient sex was also significantly associated with symptom number, as males had a lower rate of symptoms versus females (IRR 0.63, 95%CI 0.48-0.83, p=0.001). In contrast, histology (IRR 1.03, 95%CI 0.53-2.02), diagnosis year (IRR 1.01, 95%CI 0.99-1.04, p=0.35) and BMI (IRR 1.01, 95%CI 0.99-1.02) were not associated with symptom number in adjusted models. **Conclusions:** In a large nation-wide cohort, three of every 4 patients were symptomatic prior to primary AC diagnosis. Compared to the rapid time course of acute appendicitis, over 40% of this population presented with symptoms for 3+ months prior to AC diagnosis. The higher symptom burden among young patients and females supports the need for clinical providers to keep occult appendiceal tumors in the differential diagnosis of patients presenting in this manner. Research Sponsor: Dalton Family Foundation; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

825 **Poster Session**

Phase II trial of sintilimab in cancer of unknown primary.

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Background: CUP is an aggressive rare malignancy accounting for 2-4% of all cancers. Beyond frontline therapy, predominantly systemic cytotoxic chemotherapy, no clear standard of care exists. Preliminary data suggests that immune-checkpoint inhibitors have encouraging activity in patients with CUP. Prior anti-PD1 agents have demonstrated response rates of 21-23%. We studied the role of sintilimab, an anti-PD1 antibody, in CUP. **Methods:** In this phase II study (NCT05024968)10 patients (of planned 40) were enrolled prior to closure of the study due to sponsor funding. Patients were eligible if they were refractory or intolerant to one line of systemic therapy or had a contraindication for cytotoxic chemotherapy. Patients received sintilimab 200 mg IV every 3 weeks. Paired tumor biopsies were collected for biomarker analysis. The primary endpoint was confirmed objective response rate (cORR) by RECISTv1.1 assessed by independent radiology review. Key secondary endpoints were safety, disease control rate (DCR), duration of response (DOR), progression-free survival (PFS), overall survival (OS), and Quality of Life (QOL). Median PFS and OS were calculated according to the Kaplan-Meier method. Adverse events (AEs) were assessed according to CTCAE v5.0. **Results:** Between 10/2021 and 2/2023, 10 patients were enrolled and treated. Median age was 65 years and 30% of patients were female. Seven (70%) had poorly differentiated carcinoma. Patients received a median of 1 line of therapy prior to enrollment. Median follow-up was 19.8 months. All patients were evaluable for toxicity and response. The cORR was 40% (95% CI, 12.2 to 73.8); one (10%) patient with a complete response and 3 (30%) with partial responses. The median DOR was 15.7 months. Three patients (30%) had Grade ≥ 3 (no grade 4/5) treatment-related AEs including 2 patients with sepsis and 1 patient with leukocytosis. Median PFS and OS were 4.4 months (95% CI, 2.0 to 25.4) and 25.4 months (95% CI, 4.0 to 27.4), respectively. **Conclusions:** Anti-PD1 therapy with Sintilimab demonstrated significant antitumor activity for select patients with refractory CUP with safety profile consistent with other anti-PD1 agents. Ongoing correlative studies may inform the role of biomarker selection for immunotherapy in CUP. Clinical trial information: NCT05024968. Research Sponsor: None.

Peak part 1 summary: A phase 3, randomized, open-label, multicenter clinical study of bezuclastinib (CGT9486) and sunitinib combination versus sunitinib in patients with gastrointestinal stromal tumors (GIST).

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Background: After initial response to first line therapy with imatinib, GISTs commonly progress due to secondary resistance mutations in KIT. As the KIT mutation targeting profiles of bezuclastinib (type I TKI) and sunitinib are distinct, when combined they inhibit a broad spectrum of secondary KIT mutations. Herein we report extended experience of patients treated with 2nd line combination bezuclastinib and sunitinib, including response assessment and safety. **Methods:** Peak (NCT05208047), a global randomized Phase 3, open-label study, aims to evaluate efficacy and safety of bezuclastinib + sunitinib vs sunitinib in GIST pts with imatinib intolerance or resistance. In Part 1a of the 3-part study, the dose of an optimized formulation of bezuclastinib was escalated in serial cohorts to achieve a target exposure comparable to that achieved at the RP2D established in the prior Phase 1b/2a study. Part 1b evaluated the interaction between bezuclastinib + sunitinib. Key inclusion: adult with locally advanced, metastatic and/or unresectable GIST, ≥1 measurable lesion according to modified RECIST v1.1, and ECOG PS 0 to 2. Part 1 completed enrollment in Apr 2023. Based upon PK and safety, a dose of bezuclastinib 600 mg QD + sunitinib 37.5 mg QD was selected for Part 2. All Part 1 data reported herein are as of Aug 2023; updated safety and efficacy will be presented. **Results:** Part 1 has completed enrollment with 19 pts in Part 1a and 23 in 1b. In Part 1a, Cohort 1 included 5 pts starting at bezuclastinib 300 mg QD + sunitinib 37.5 mg QD; Cohort 2 included 14 pts at bezuclastinib 600 mg QD + sunitinib 37.5 mg QD. Pt characteristics in Part 1a+1b: median age - 60 yrs (range: 33-77); 81% men; 98% ECOG PS 0-1; 98% metastatic and 2% locally advanced. The majority of TEAEs were low grade and reversible with low rate (38%) of Grade 3+ events. There were limited (24%) dose reductions and infrequent (n=2) discontinuations due to TEAEs. Three pts experienced serious AEs possibly associated with study medications. In pts evaluable for response (received ≥1 cycle of study treatment and had assessments at baseline and post-baseline [n=40]), the objective response rate (ORR) was 20% (6 confirmed PRs, 2 unconfirmed PRs). Data remain immature to estimate median PFS for Part 1 pts. In a subset of pts with prior imatinib only (n=6), the closest approximation for pts enrolling in Part 2, the ORR was 33% (2 confirmed PRs). The majority of 2nd-line pts remain on treatment past 12 months. **Conclusions:** Data from Peak Part 1 show an encouraging safety and tolerability profile generally consistent with published sunitinib monotherapy experience. ORR in evaluable pts from Part 1 was 20%; ORR in 2nd line pts was 33%. Part 2 of the Peak study is actively enrolling pts globally at the selected dose of bezuclastinib 600 mg QD + sunitinib 37.5 mg QD versus sunitinib 37.5 mg QD. Clinical trial information: NCT05208047. Research Sponsor: Cogent Biosciences.

A phase 1/2 study evaluating the safety and efficacy of autologous TAC T cells in subjects with claudin 18.2+ advanced solid tumors.

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Background: Claudin18.2 (CLDN18.2) is a tight junction protein found in gastric epithelial cells. It can become abnormally expressed in gastric cancer and other solid tumors, rendering it a candidate for targeted therapy. There are no approved therapeutic agents against CLDN18.2. The T cell antigen coupler (TAC) technology modifies T cells *ex vivo*, allowing cytotoxicity of tumor cells by co-opting the natural T cell receptor. TAC T cells demonstrate a safer profile than chimeric antigen receptor T cells. TAC01-CLDN18.2 is an autologous T-cell product comprising T cells expressing CLDN18.2 TAC. **Methods:** In this first-in-human study (NCT05862324), subjects will undergo leukapheresis and may receive bridging anticancer therapy during cell manufacturing. Prior to TAC01-CLDN18.2 infusion, subjects will undergo lymphodepletion chemotherapy. In Phase I, TAC01-CLDN18.2 will be administered at increasing doses (3 cohorts) in adult subjects with ≥2 lines of prior therapy (1 for subjects with pancreatic ductal adenocarcinoma (PDAC)) using the classic 3+3 dose escalation study design. CLDN18.2 expression levels will be determined centrally using a validated clinical trial assay. Dose-limiting toxicities (DLTs) will be assessed up to 28 days after the infusion. A second dose may be administered according to preidentified clinical and safety criteria. In Phase II, dose expansion groups will evaluate the efficacy, safety, and pharmacokinetics of the optimal TAC01-CLDN18.2 dose, with the option of redosing. Indications will include gastric and esophageal adenocarcinoma (group A), PDAC (group B) and ovarian and non-small cell lung cancer (group C) in subjects with <4 lines of prior therapy. **Results:** The first two dose cohorts of Phase I have been completed, with no reported dose-limiting toxicities (DLT). All of the TAC-related adverse events were low-grade. No cytokine release syndrome was reported. One subject in cohort 2 experienced a grade 1 neurotoxicity, which resolved the same day without intervention. Six subjects reported a total of 6 serious adverse events, none related to TAC01-CLDN18.2. An 83.3% disease control rate was observed at first tumor assessment (Day 29). One subject in cohort 1 with heavily pre-treated PDAC (3 prior lines) with high CLDN18.2 expression has an ongoing, confirmed partial response. The subject received a second dose and is still on treatment (6 months). **Conclusions:** Treatment with the lower doses of TAC01-CLDN18.2 is safe and shows promising clinical activity in a heavily pre-treated cancer population. Treatment of cohort 3 has begun. Clinical trial information: NCT05862324. Research Sponsor: Triumvira Immunologics.

829 **Poster Session**

AGX101: A TM4SF1-directed tubulin inhibitor conjugate in ongoing first-in-human trial including GI cancers.

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Background: TM4SF1 (Transmembrane-4 L-Six-Family-Member-1) is an endothelial marker with critical roles in angiogenesis, as well as a tumor cell antigen that contributes significantly to invasion and metastasis. TM4SF1 is upregulated 20-fold in angiogenic tumor vascular endothelium compared to normal vasculature endothelium and exhibits a unique nuclear internalization pathway. AGX101 is a novel tubulin inhibitor conjugate specifically directed against TM4SF1, delivering a potent maytansinoid payload directly to the nucleus of cells within the tumor microenvironment. Delivery to both the vascular and tumor cell compartments of the tumor results in three mechanisms of action (MoAs): (1) activation of tumor immune surveillance, (2) tumor blood supply deprivation, and (3) direct tumor cell killing. **Methods:** TM4SF1 expression was assessed using immunohistochemistry. Safety and pharmacokinetics of AGX101 were evaluated in non-human primates (NHP), with escalating doses to determine the highest non-severely toxic dose (HNSTD). Efficacy studies were also conducted in mouse models, enabling assessment of the minimum effective dose (MED) needed to engage each of the three MoAs. The combination of HNSTD and MED enables calculation of a therapeutic index (TI). The potential for synergy with immune checkpoint inhibitors (ICIs) was also investigated. A first-in-human study of AGX101 in patients with advanced solid tumors with the primary objective of safety and dose-limiting toxicities (DLTs) has initiated. **Results:** Notable cancers with high TM4SF1 scoring intensity include pancreatic adenocarcinoma, hepatocellular carcinoma, cholangiocarcinoma, and gastric cancer. In esophageal cancer, 20% of patients have a TM4SF1 copy number amplification, and these have worse prognosis. In NHP, AGX101 exhibited a favorable safety profile. In preclinical efficacy studies, monotherapy demonstrated robust efficacy through each of the three MoAs in syngeneic models in mice including CT26 colon carcinoma, and human tumor xenograft models including MIA PaCa-2 pancreatic cancer. Measured by exposure, TI was large. In syngeneic mouse models including CT26, the effects of ICIs were potentiated, suggesting a potential synergistic approach in cancer therapy. The first two dose levels of AGX101 monotherapy in humans has cleared without DLTs. Three patients with pancreatic adenocarcinoma have been treated thus far. **Conclusions:** AGX101, targeting the TM4SF1 antigen, represents a promising new approach in cancer therapy. The preclinical data suggest that AGX101 could provide a significant therapeutic benefit by novel and differentiated mechanisms of action, namely selectively targeting the tumor vasculature and potentiating immune-based therapies. Further clinical development of AGX101 is ongoing and initial outcome data will become available by the conference. Clinical trial information: NCT06440005.
Research Sponsor: Angiex, Inc.

A 20-year population-wide cohort study on association of aspirin and risk of gastrointestinal and non-gastrointestinal cancer.

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Background: Aspirin has been shown to reduce the risk of various gastrointestinal (GI) and non-GI cancers[1,2]. However, little was known about at what age aspirin should be started for maximal chemoprotective better on GI cancer. Furthermore, the randomised controlled trial on aspirin for primary prevention use in healthy elderly (ASPREE) showed no association between aspirin and cancer incidence despite the limitation of short follow-up duration[3]. The current study aims to investigate the 20-year risk of cancer using aspirin in a territory-wide Hong Kong population cohort. **Methods:** The study included all aspirin users from 2000 to 2019, and non-aspirin users matched by age and sex at a ratio of 1:2. Enrolled subjects with a history of cancer at enrolment, cancer incidence or death within 6 months were excluded. The incidence of individual GI and non-GI cancer was presented as the primary outcome. Baseline characteristics between aspirin and non-aspirin users were adjusted in the survival analysis by inverse probability of treatment weighting (IPTW). The fine-grey model has been used to address bias from competing risk of death. The sub-distribution hazard ratio was presented for the association of aspirin use and risk of GI and non-GI cancers. **Results:** The current study included 538,147 aspirin users and 968,378 non-users with a mean age of 64.8 years. A total number of 36,683 cases of GI cancer (2.4%) and 47,196 cases of non-GI cancers (3.1%) were observed. Aspirin was associated with a lower risk of several common individual GI cancers, including colorectal cancer (SHR 0.78, 95% CI 0.76-0.81), liver cancer (SHR 0.67, 95% CI 0.64-0.70), stomach cancer (SHR 0.79, 95% CI 0.75-0.84) and pancreatic cancer (SHR 0.85, 95% CI 0.79-0.91), but not oesophageal cancer. On the other hand, aspirin was associated with a lower risk of prostate cancer (SHR 0.95, 95% CI 0.91-1.00) and breast cancer (SHR 0.76, 95% CI 0.73-0.79), but not lung cancer and kidney cancer. In overall, aspirin was associated with a 24% lower risk of GI cancers (SHR 0.76, 95% CI 0.74-0.78) and a 3% lower risk of non-GI cancers (SHR 0.97, 95% CI 0.95-0.99). **Conclusions:** Aspirin was associated with a lower risk of most GI cancers, including colorectal cancer, liver cancer, stomach cancer and pancreatic cancer, but not most non-GI cancers. In general, results on the effect of aspirin on GI cancer prevention were consistent with the previously presented 10-year cohort. 1. Bosetti C, Santucci C, Gallus S, Martinetti M, La Vecchia C. Aspirin and the Risk of Colorectal and Other Digestive Tract Cancers: An Updated Meta-analysis through 2019. Ann Oncol. 2020;31(5):558-568. 2. Santucci C, Gallus S, Martinetti M, La Vecchia C, Bosetti C. Aspirin and the risk of nondigestive tract cancers: An updated meta-analysis to 2019. Int J Cancer. 2021;148(6):1372-1382. 3. McNeil JJ, Gibbs P, Orchard SG, et al. Effect of Aspirin on Cancer Incidence and Mortality in Older Adults. J Natl Cancer Inst. 2021;113(3):258-265. Research Sponsor: Health and Medical Research Fund, by Health Bureau Hong Kong.

831 **Poster Session**

Uses of VEGF inhibitor bevacizumab in appendiceal adenocarcinoma.

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Background: Systemic treatment of metastatic appendiceal adenocarcinoma (AA) has been challenging due to a lack of well-established AA chemotherapy regimens and historical reliance on colorectal cancer (CRC) protocols. In the United States, but less commonly in Europe, bevacizumab is frequently incorporated in treatment of AA given its approved use for CRC. However, previously there have been no studies evaluating if there is a survival benefit attributable to bevacizumab in patients with AA. **Methods:** The Palantir Foundry system was used to extract data from the MD Anderson Cancer Center (MDA) electronic medical record for patients with biopsy-proven AA who received chemotherapy at MDA from 2015 to 2024. Only patients with complete staging and histopathology data were included in the analysis. **Results:** We identified 203 patients with AA who received chemotherapy at MDA and had complete tumor data. Of these, 43% (n=87) received bevacizumab at some point in their treatment. Patients who received bevacizumab were demographically similar to patients who received non-bevacizumab chemotherapy in terms of age at diagnosis, race, and sex; mucinous histology was more frequent in the bevacizumab cohort (52%, n=45 vs. 39%, n=45 in non-bevacizumab cohort). The 87 patients who received bevacizumab were treated with 1-6 lines of therapy per patient (mean = 1.57). The most frequently used bevacizumab-containing regimens were FOLFOX + bevacizumab (34%, n=46), FOLRIRI + bevacizumab (29%, n=40), 5-FU + bevacizumab (15%, n=21), and atezolizumab + bevacizumab (11%, n=15), with less common irinotecan + bevacizumab, TAS-102 + bevacizumab, and FOLFOXIRI + bevacizumab. Three regimens containing bevacizumab were discontinued due to complications: one instance of hand-foot syndrome attributed to xeloda, one instance of cardiomyopathy attributed to immunotherapy, and one bowel perforation with abdominal wall abscess attributed to bevacizumab. To determine if bevacizumab was associated with overall survival (OS) benefit we performed Kaplan-Meier analysis, with OS calculated from start of first line chemotherapy to death, in patients with metastatic AA who received either 5-FU doublet therapy alone or 5-FU doublet therapy with bevacizumab. Patients who received bevacizumab trended toward better OS (median 52 vs 25 months for doublet therapy alone, HR: 0.53, p= 0.059). The effect of bevacizumab was particularly notable in patients with signet ring cell histology; where bevacizumab use was associated with significantly longer OS (median OS 50 vs 14 months for doublet therapy alone, HR: 0.24, p= 0.034). **Conclusions:** Bevacizumab is frequently added to cytotoxic chemotherapy in the treatment of AA. These retrospective data suggest an OS benefit to adding bevacizumab to doublet chemotherapy, particularly in the case of signet ring cell histology. Research Sponsor: None.

A comparative study of dosage strategies in HIPEC: BSA-based vs fixed dose methods.

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Background: Peritoneal metastases commonly occur in ovarian, gastric, and colorectal cancers, significantly contributing to patient morbidity and mortality. Hyperthermic Intraperitoneal Chemotherapy (HIPEC) is a promising approach for managing peritoneal surface cancers. This procedure involves cytoreductive surgery to remove visible tumors, followed by the perfusion of heated chemotherapy (41-42°C) directly into the peritoneal cavity. HIPEC addresses the limitation of systemic chemotherapy, which often fails to penetrate the peritoneal cavity effectively. HIPEC dosing may either be a flat-fixed dose or be calculated based on Body Surface Area (BSA). BSA can be determined using three methods: actual BSA (based on weight and height), ideal BSA (based on sex and height), and adjusted BSA (considering lean and fat mass). **Methods:** A retrospective chart review was conducted on 151 metastatic cancer patients at UVMMC. Patient BMI, weight, and height at diagnosis were collected to calculate BSA values (actual, adjusted, and ideal) for estimating a hypothetical chemotherapy dosage. Statistical analysis using SPSS involved linear mixed models and paired t-tests to compare Mitomycin-C dosages between a fixed dose (40 mg) and a BSA-derived dose (30 mg/m^2). **Results:** The average dosage of Mitomycin-C differed significantly depending on whether the dose was calculated using actual, adjusted, or ideal BSA. The largest discrepancy in dosage was observed between the actual BSA method and the flat-fixed dose, with an average difference of 16 mg. The smallest difference in dose was between the adjusted and ideal BSA methods. When stratifying by BMI, the greatest variation in Mitomycin-C dosage was found among overweight and obese patients (BMI $\geq$ 25). For these patients, the average dosage difference between the flat fixed dose and the actual BSA dose was 20 mg. In comparison, in patients with a BMI under 18.5 (underweight) the actual BSA dose was comparable to the fixed dose. **Conclusions:** Different institutions utilize a variety of dosing strategies for the HIPEC procedure. Differences in BSA calculation methods and perfusate volumes (literature values of 2, 3 and 4 L is most common) contribute to the complexity of establishing standardized dosing. The significant dosing variations observed, especially in overweight and obese patients, are of concern, as 70% of the U.S. population has a BMI $\geq$ 25. Our study demonstrates significant variability in HIPEC chemotherapy dosing, especially among patients with higher BMI. The discrepancy between BSA-based methods and flat-fixed doses may impact treatment efficacy and toxicity. Standardizing dosing approaches is important to ensure consistent therapeutic outcomes and patient safety. Research Sponsor: None.

Personalised use of repurposed non-anticancer drugs in colorectal cancer based on up-regulated signalling pathways: A real world cohort feasibility study.

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Background: Targeting signaling pathways in colorectal cancers is a common therapeutic strategy. In addition to targeted drugs, non- anticancer drugs approved for use in other health conditions can inhibit key pathways. De novo tumor transcriptome analysis can identify patient-specific up- and down-regulated transcripts, based on which the corresponding repurposed drugs can be used in synergistic combinations with standard of care (SoC) regimens. **Methods:** This retrospective study was conducted on a real-world cohort of 459 colorectal cancer cases (218 females, 241 males; median age 56 years) who had taken Exacta multi-omics tumor profiling for selection of personalized treatments. Transcriptome analysis was performed as part of the evaluations to determine up and down-regulated gene transcripts for selecting targeted, endocrine, and cytotoxic anticancer agents as well as non-anticancer (repurposed) drugs and phytochemicals that could be used for tandem targeting of the dysregulated signaling pathways, as indicated in the table below. **Results:** Transcriptome analysis of this cohort revealed dysregulation of ≥1 pathway in 391 patients (85.2%). MAPKs (n = 244, 53.2%) and MMPs (n = 210, 45.8%) were most frequently upregulated. Celecoxib (n = 292, 63.6%), Atorvastatin (n = 255, 55.6%), and Doxycycline (n = 208, 45.3%) were the most frequently indicated repurposed drugs for targeting the identified dysregulated pathways. Potential indications for ≥1 repurposed drugs were seen in 353 patients (76.9%). **Conclusions:** Several cellular signaling pathways are known or potential therapeutic targets in cancers. The present study indicates that it may be possible to use multi-omics analyses to create personalized treatment strategies using a synergistic combination of repurposed drugs to potentiate the action of SoC systemic anticancer agents in colorectal cancer. Research Sponsor: None.

Biomarkers for non anti-cancer drugs selection.

Drug	Biomarker/Gene Expression
Aspirin	PTGS2 (COX2)
Atorvastatin	HMGCR, MAPK
Bromelain	PTGS2 (COX2)
Celecoxib	PTGS2 (COX2), FZD, WNT, MAPK
Chloroquine	HMGB1
Curcumin	MMP, BIRC5, BCL2
Doxycycline	MMP
Metformin	MMP, BCL2, MAPK
Quercetin	FZD, WNT
Resveratrol	MMP, BCL2, PCNA, PTGS2 (COX2)
Vitamin C	SLC2A1 (GLUT1)
Artesunate	FZD, WNT, MMP, BCL2
Cannabidiol	MMP
Epigallocatechin gallate	MMP, MAPK
Mebendazole	XIAP, MMP, MAPK

CK2 expression gastrointestinal cancer: Exploring clinical characteristics and cancer genetics.

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Background: The expression of protein kinase CK2 subunits (CSNK2A1, CSNK2A2, CSNK2A3, and CSNK2B) in gastrointestinal cancers has not been fully characterized. In this study, we seek to evaluate how changes in CK2 subunit expression are associated with GI cancer clinical characteristics and genetics. **Methods:** Data analysis was performed with data from The Cancer Genome Atlas (TCGA). R version 4.2.2 was used for analysis. We provide an analysis of several cancer cohorts looking at CK2 subunit expression, as well as cancer mutations, copy number alterations, and clinical characteristics. **Results:** From the TCGA datasets containing mRNA expression data, we assessed hepatocellular carcinoma (n=361), colorectal adenocarcinoma (n=339), gastric adenocarcinoma (n=306) and diffuse type gastric adenocarcinoma (n=69), pancreatic adenocarcinoma (n=169), esophageal adenocarcinoma (n=89) and squamous cell carcinoma (n=95), and cholangiocarcinoma (n=36). We explored associations between CK2 subunit expression and GI cancer-specific known driver mutations, of which we report the following significant findings ($p < 0.05$ for all). Mutations in TP53 were associated with increased expression of CSNK2A1 in hepatocellular carcinoma and stomach adenocarcinoma, increased CSNK2A2 in colorectal adenocarcinoma, and CSNK2B in colorectal adenocarcinoma, pancreatic adenocarcinoma, and stomach adenocarcinoma. Mutations in KRAS were associated with increased expression of CSNK2A1 and CSNK2B in pancreatic adenocarcinoma. Mutations in BRAF were associated with decreased expression of CSNK2A2 and CSNK2B in colorectal adenocarcinoma. We also noted significant changes in CSNK2 subunit expression when looking at mutations and copy number alterations in signaling pathways such as Wnt, NOTCH, PI3K, TGFβ, and more. We also explored associations between CK2 subunit expression and GI cancer clinical characteristics, of which we report the following significant findings ($p < 0.05$ for all). In colorectal adenocarcinoma, increased CSNK2A2 expression was associated with lymph node invasion and advanced clinical stage. In hepatocellular carcinoma, increased CSNK2A1 and CSNK2A3 expression was associated with greater tumor grade, tumor size, and more advanced clinical staging. Increased CSNK2A2 expression was associated with greater tumor size, more advanced clinical staging, and vascular invasion. Increased CS2NKB expression was associated with increased tumor grade, size, and vascular invasion. In pancreatic adenocarcinoma, increased CSNK2A3 expression was associated with a worse response to the first treatment course. **Conclusions:** Our results suggest that changes in protein kinase CK2 expression are associated with different cancer clinical characteristics as well as cancer-specific mutations and signaling pathways. Research Sponsor: None.

835 **Poster Session**

Characterization of the tumor immune microenvironment (TIME) and somatic landscape in gastrointestinal (GI) malignancies with *MTAP* deletions (del).

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Background: PRMT5 is a synthetic lethality target in patients (pts) with *MTAP* del and early phase trials are underway with PRMT5 inhibitors. Additionally, *MTAP* del are associated with a less immunogenic TIME and reduced efficacy of immunotherapy, but research has primarily not been focused on GI malignancies. Thus, we investigated the TIME and somatic landscape in GI malignancies with *MTAP* del. **Methods:** From the Tempus Database, we retrospectively analyzed de-identified next-generation sequencing data from pts across GI malignancies, including pancreatic (n=11,217), gastroesophageal (GEJ, n=5,803), cholangiocarcinoma (CCA, n=3,244), and colorectal (CRC, n=17,537) cancers. Tumors were sequenced with the Tempus xT DNA (648-gene panel) and xR RNA assays. *MTAP* del were defined as two-copy losses. Somatic alterations (alt), immune cell infiltration predicted from gene expression patterns, PD-L1 from IHC, TMB, and MSI were evaluated. Chi-squared/Fisher's Exact tests or Kruskal-Wallis tests were used to assess statistical significance ($p<0.05$, $q<0.05$ for false discovery rate correction for multiple testing). **Results:** *MTAP* del were identified in 14.8%, 11.3%, 0.9%, and 7.3% of pancreatic, CCA, CRC, and GEJ cancers, respectively. Of these, 98/93%, 99/91%, 92/89%, and 98/89% had co-occurring *CDKN2A/B* loss. In pancreatic, CCA, and CRC pts, *MTAP* del was associated with a reduced proportion of B cells and CD4 T cells, and there were higher percentages of macrophages vs pts with *MTAP* WT status ($p<0.001$ for all). Reductions in proportion of CD8 T cells were also associated with *MTAP* del in pancreatic and CCA pts ($p<0.001$ for both). Lower TMB ($p=0.036$) and MSI-H status ($p=0.009$) were found in CRC pts with *MTAP* del. GEJ with *MTAP* del also exhibited significantly lower MSI-H status than *MTAP* WT GEJ ($p=0.014$). *SMAD4 alterations*, a marker of reduced immune infiltrates, were more prevalent in pts with *MTAP* del across GI malignancies ($q<0.005$). In the CCA cohort, there was a higher percentage of *BRAF* alt and FGFR2 fusions in pts with *MTAP* loss ($q<0.001$, $q=0.028$), while *KRAS* alt were higher in pancreatic cases with *MTAP* loss ($q<0.001$). **Conclusions:** This is the largest analysis of the TIME and somatic landscape of *MTAP* loss across GI malignancies. In pts with *MTAP* del and pancreatic cancer, CCA, and CRC, we observed a less immunogenic TIME pattern, indicating the evaluation of immunotherapy implications in these GI malignancies with *MTAP* del is warranted. Our findings are hypothesis-generating, providing further rationale to study synthetic lethality and novel combinatorial therapeutic strategies in GI malignancies with *MTAP* del. Research Sponsor: Tempus AI, Inc.

Short interval circulating tumor DNA (ctDNA) kinetics as a predictor of tumor response in patients with gastrointestinal (GI) cancer receiving immune checkpoint inhibitor (ICI)-based treatment.

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Background: Circulating tumor DNA (ctDNA) has a short half-life (<2 hours) which may permit real-time monitoring of tumor status. This single-institution study aimed to assess the feasibility of rapid treatment response evaluation through serial short-interval ctDNA testing. **Methods:** Patients with gastrointestinal (GI) cancer undergoing immune checkpoint inhibitor (ICI)-based therapy were included. A personalized, tumor-informed ctDNA assay (Signatera, Natera, Inc.) was used in this study. We collected samples to measure the baseline ctDNA levels on cycle 1, day 1 (C1D1) of treatment or within 14 days prior to C1D1. A second ctDNA sample was obtained after 14-21 days depending on the cycle length of the treatment regimen employed (on cycle 2, day 1 [C2D1]). Changes in ctDNA levels between baseline and C2D1 were recorded, as well as the tumor response at the first radiographic assessment. Patients achieving partial response (PR) or better, or stable disease (SD) with tumor shrinkage supporting continuation of same treatment were considered responders. **Results:** The study cohort consisted of 14 patients aged 36-89 years (median age 66); 5 patients were female. Tumor types included advanced colorectal (n=5) and gastroesophageal (n=9) cancers. Treatments consisted of pembrolizumab alone (n=5), ICI + chemotherapy combinations (n=8), and regorafenib plus nivolumab (n=1). Baseline ctDNA levels were obtained on C1D1 in 9 patients, with the remaining 5 patients having baseline measurements within 14 days before C1D1. All patients had the second ctDNA level drawn on C2D1. The median interval between C1D1 and the first radiographic response assessment was 61 days (range, 45-84). A reduction in ctDNA levels by 50% or more (range: 66-100%) was observed in 9 patients (64%) after one cycle of treatment. All 9 patients were classified as responders at the time of first radiographic assessment: 7 with PR and 2 with SD but significant tumor shrinkage leading to continuation of the same treatment. In 3 patients (21%), ctDNA levels increased (range: 20-500%), correlating with progressive disease (PD) on first assessment scan. Two patients showed increased ctDNA levels of 82% and 22% after one cycle of treatment but achieved PR and SD, respectively. In both patients, however, the 3rd ctDNA level drawn before the first assessment scan dropped by >90%. Overall, short-interval ctDNA kinetics correlated with treatment response in 12 of 14 patients (86%), with a binomial test $p = 0.0065$. **Conclusions:** Short-interval ctDNA kinetics demonstrates a strong correlation with radiographic tumor response and clinical benefit in GI cancer patients receiving ICI-based treatments, providing early signal of tumor response. This approach has the potential to guide treatment decisions if validated in larger prospective studies. Research Sponsor: None.

Use of gene expression profiling to examine association between MUC1 and claudin 18 in gastrointestinal cancers: Implications for targeted therapies.

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Background: Gastrointestinal tract cancers exhibit considerable molecular heterogeneity, which requires a detailed understanding of their distinct genetic profiles. Understanding the gene expression of other genes along with Claudin 18 (CLDN18) could provide insights into their interactions. **Methods:** We conducted a retrospective analysis of NGS profiling on 1,043 GI cancer tissue samples from three cohorts: Upper and Lower Gastrointestinal Cancer Cohort (GICC; n=651), Pancreaticobiliary Cancer Cohort (PBCC; n=316), and Hepatocellular Carcinoma Cohort (HCC; n=76). Immunohistochemistry data for PD-L1, Her2 and MMR proteins, NGS based Tumor mutation burden (TMB) and gene expression profiling (GEP) of 20,802 genes including CLDN18 was examined in a subset by targeted transcriptome profiling. We explored associations of these biomarkers with expression of CLDN18. **Results:** In GICC, frequent mutations were seen in TP53 (68%; 442), APC (33.5%; 218) and KRAS (33.3%; 217). PBCC, showed mutations in TP53 (62%; 197), KRAS (47%; 149) and CDKN2A (16%; 51). HCC showed mutations in TP53 (42%; 32), ARID1A (18%; 14) and KRAS (9%; 7). Amplifications were predominantly observed in MYC (13%; 120), ERBB2 (5%; 49) and CCND1 (5%; 44). Fusions were infrequent. All MSI-H cases (11/462; 2.4%) were observed in GICC. High TMB (≥10 muts/mb) was seen in 23% (111/481), stratified as 28% (90/318) in GICC, 13% (18/139) in PBCC and 10% (3/30) in HCC. Out of 321 samples analyzed for gene expression profiling, CLDN18 was significantly upregulated in 65 (20%) samples. The expression of CLDN18 did not show statistically significant correlation with Her2 (n=184), PD-L1 (n=231), TMB (n=228), or MSI (n=229); ($p>0.05$). In subset where both CLDN18 and MUC1 expression were analyzed (n=118), all the CLDN18 positive tumors showed overexpression of MUC1. This finding was statistically significant ($p<0.05$). Increased expression of MUC1 promotes cancer cell mobility promoting metastases. Expression of MUC1 can play an important role in the development of resistance to chemotherapy. MUC1 expression in cancer cells is associated with avoidance of immune defenses. Thus, this finding of CLDN18 and MUC1 association is especially relevant as the CLDN18 targeted therapies may be the potential avenue for the aggressive and chemo-resistant MUC1 expressing tumors. **Conclusions:** This study identifies potential association between MUC1 and CLDN18. GEP could enhance biomarker detection and support personalized treatment approaches. Future research should integrate these findings with clinical data to improve precision in GI cancer care. Research Sponsor: None.

Clinical validation of the Northstar Response: A novel quantitative methylation ctDNA monitoring assay for advanced GI cancer treatment response.

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Background: Circulating tumor DNA (ctDNA) is a promising tool for monitoring treatment response but is challenged by pan-cancer quantification and consistent outcome correlations. Northstar Response (NSR) quantifies >500 cancer specific methylated loci, offering a reflection of disease burden and clinical response. **Methods:** We present a prospective observational study to evaluate the clinical utility of NSR. Patients with advanced GI cancers had serial ctDNA assessed at baseline and during systemic treatment with correlations to radiographic or clinical response. Tumor methylation scores (TMS) were expressed as Log10 values and a mixed-effects model evaluated clinical assessment correlations. **Results:** 73 patients with locally advanced or metastatic GI cancers were analyzed. Among these, 35 (47.9%) had at least four ctDNA assays and all had radiographic or clinical response assessments. The median age was 66 years (± 8.5), with most being male (67.1%) and White (74.0%). Table 1 shows the case distribution. Baseline TMS had significant variation (p=0.023), but was measurable across all tumor types. CRC exhibited the highest mean TMS-log10 while biliary and pancreatic cancers had the lowest. Cases with both primary and metastatic lesions had a higher mean TMS-log10 than cases with either primary tumors alone or metastatic lesions alone (3.4 vs. 3.0). Compared to stable and responsive disease, TMS changes correlated with progressive disease, with a coefficient of 0.48 at the third collection time point (p=0.05). However, 23/73 (32%) cases were without an on-treatment TMS timepoint (OTT) due to rapid progression (average time to death of 24 vs. 103 days for patients with OTT). In exploratory modeling of TMS relative to RECIST-like treatment response for those with OTT, NSR evaluations were associated with response assessments (p<0.04, Fisher's Exact Test). **Conclusions:** TMS was consistently measurable but varied across tumor types and correlated with disease burden. TMS changes appeared associated with response assessments when accounting for rapid progressors. Additional analyses are ongoing and will be presented. Research Sponsor: BillionToOne, Inc.; University of Florida Health Cancer Center; P30CA247796.

Baseline study cohort characteristics.

Characteristic	HCC N = 25 (34%)	BTC N = 12 (16%)	EGA N = 12 (16%)	CRC N = 11 (15%)	PDAC N = 8 (11%)	Other GI N = 5 (7%)	p-value
Age Median (25% to 75%)	67 (63 - 76)	68.5 (62.3 – 75.5)	73.5 (62.5 – 75.3)	54 (39.5 – 60)	67.5 (55.5 – 76.3)	60 (57 – 63)	0.008
Baseline TMS Log10 Median (25% - 75%)	3.2 (2.3 – 3.9)	2.5 (2.1 – 3.4)	3.2 (2.2 – 3.8)	4.4 (3.5 – 5)	2.5 (1.9 – 3.3)	4 (3.4 – 5.2)	0.023
Primary tumor present	12 (48%)	8 (66.7%)	12 (100%)	6 (54.5%)	7 (87.5%)	4 (80%)	0.015
First line therapy vs. not	23 (92%)	5 (41.7%)	5 (41.7%)	6 (54.5%)	3 (37.5%)	5 (100%)	0.001
Progressive disease	16 (64%)	4 (33.3%)	9 (75%)	5 (45.5%)	3 (37.5%)	4 (80%)	0.2
Death	13 (52%)	3 (25%)	9 (75%)	7 (63.6%)	3 (37.5%)	2 (40%)	0.2

Kruskal-Wallis rank sum test; Fisher's exact test.

Universal screening for mismatch repair deficiency in gastrointestinal cancers in a reference center in Mexico.

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Background: Gastrointestinal (GI) cancers have a poor prognosis. With the availability of immunotherapy for tumors with deficient mismatch repair (dMMR), identifying these patients has become increasingly important. Universal dMMR screening across all GI cancers is uncommon, and its prevalence has not been well studied prospectively. In Mexico, the prevalence of dMMR is unknown, even though GI cancers account for a large portion of new cancer cases and cancer-related deaths. Our objective was to prospectively determine the prevalence of dMMR in GI cancers at a referral center in Mexico. **Methods:** Prospective universal screening for dMMR began in March 2024. Primary tumor sites included were esophagus, stomach, pancreas, biliary tract, small intestine, colon, and rectum. Histologies included were adenocarcinoma, squamous cell, or mixed. All consecutive patients were included regardless of age, clinical stage or medical history. MMR status was determined by immunohistochemistry, analyzing expression of MLH1, PMS2, MSH2, and MSH6. This abstract presents the results from the first six months of this universal screening. **Results:** A total of 161 GI cancers were included (see Table). Prevalence of dMMR was 9.3% (n=15). The highest prevalence was in gastric cancer (13.3%, 2/15), followed by colon (12.1%, 7/58), pancreatic (11.1%, 4/36), and bile duct (8.3%, 2/24) cancer. MMR status was not performed in 9.3% (n=15) of cases. Combined deficiency was found in 60% (9/15) for MLH1/PMS2 and 20% (3/15) for MSH2/MSH6. Isolated deficiency occurred in 20% (3/15) for PMS2 and 13.3% (2/15) for MSH2. No difference in age or clinical stage was observed between MMR status groups. **Conclusions:** The high prevalence of dMMR, along with the poor prognosis and availability of immunotherapy based treatments, supports the need for universal screening. The high prevalence of dMMR in pancreatobiliary cancers was notable. Implementation efforts in our institution are essential to obtain MMR status in all cases. Research Sponsor: None.

Characteristics of population.

		pMMR n=131 (%)	dMMR n=15 (%)	Not performed n=15 (%)	Total n=161	p value
Gender	Male	63 (48.1)	10 (66.7)	7 (46.7)	80 (49.7)	0.38
	Female	68 (51.9)	5 (33.3)	8 (53.3)	81 (50.3)	
Age	Median (IQR)	65 (54 -72)	52 (43 -80)	65 (57 -71)	65 (53 -72)	0.52
Histology	Adenocarcinoma	130 (99.2)	15 (100.0)	14 (93.3)	159 (98.8)	N/A
	Mixed	1 (0.8)	0 (0.0)	0 (0.0)	1 (0.6)	
	Squamous Cell	0 (0.0)	0 (0.0)	1 (6.7)	1 (0.6)	
Clinical Stage	Unknown	12 (9.2)	1 (6.7)	1 (6.7)	14 (8.7)	0.49
	Localized	24 (18.3)	2 (13.3)	2 (13.3)	28 (17.4)	
	Locally advanced	40 (30.5)	8 (53.3)	3 (20.0)	51 (31.7)	
	Metastatic	55 (42.0)	4 (26.7)	9 (60.0)	68 (42.2)	
Tumor site	Esophageal	9 (6.9)	0 (0.0)	1 (6.7)	10 (6.2)	N/A
	Gastric	13 (9.9)	2 (13.3)	0 (0.0)	15 (9.3)	
	Small B.	1 (0.8)	0 (0.0)	0 (0.0)	1 (0.6)	
	Bile duct	14 (10.7)	2 (13.3)	8 (53.3)	24 (14.9)	
	Pancreas	29 (22.1)	4 (26.7)	3 (20.0)	36 (22.4)	
	Colon	50 (38.2)	7 (46.7)	1 (6.7)	58 (36.0)	
	Rectal	15 (11.5)	0 (0.0)	2 (13.3)	17 (10.6)	

841 **Poster Session**

Can an alteration of the gut microbiome increase risk of gastrointestinal malignancies? Insights from the National Inpatient Sample.

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Background: The gut microbiome is composed of trillions of microorganisms, inhabiting the lining of the gastrointestinal (GI) tract. The gut microbiome plays multiple roles in upholding human health including maintaining metabolism and regulating immune homeostasis. Dysregulation of the gut flora is known as gut dysbiosis. Studies have shown that gut dysbiosis is linked to the development of malignancies such as colorectal cancer. The purpose of this study is to assess the correlation between an altered gut flora and development of GI malignancies. **Methods:** The National Inpatient Sample was queried between 2016-2020 for all adult patients with altered gut flora, defined by history of gastrointestinal/bariatric surgeries, history of clostridium difficile infection, long-term antibiotic use, and inflammatory bowel disease. Rates of various cancers were compared between this cohort and all other patients utilizing chi-squared tests. Binary logistic regressions were utilized to calculate odds ratios of developing the various cancers, adjusted for age, sex, Charlson Comorbidity Index (CCI), and smoking status. **Results:** A total of 7,758,168 hospitalizations with altered gut flora were identified. Esophageal cancer (aOR 3.16, 95% CI 3.08 - 3.24) and colorectal cancer (aOR 3.15, 95% CI 3.11 - 3.18) were observed to have the highest odds of occurring in patients with altered gut flora. Gastric, hepatobiliary, pancreatic, and neuroendocrine cancers also all had significantly higher odds of occurring in patients with altered gut flora. **Conclusions:** Our study shows that conditions altering the gut flora are associated with increased odds of malignancies of the GI tract. This suggests that the gut microbiome may play an intricate role in regulation of tumor cell proliferation. Optimization of gut health and restoration of gut flora may prove beneficial to patients at high risk for malignancies of the GI tract. Research Sponsor: None.

	Altered Gut Flora (%)	Intact Gut Flora (%)	Adjusted Odds Ratio (CI)	p-value
Esophageal	0.4	0.1	3.16 (3.08 - 3.24)	**<.001**
Gastric	0.3	0.1	1.67 (1.64-1.69)	**<.001**
Colorectal	2.5	0.6	3.15 (3.11 - 3.18)	**<.001**
Hepatobiliary	0.4	0.3	1.04 (1.02 - 1.07)	**<.001**
Pancreatic	0.6	0.3	1.34 (1.31 - 1.37)	**<.001**
Neuroendocrine	0.2	0.1	1.38 (1.33 - 1.44)	**<.001**

Comparing outcomes for early and late drainage of malignant ascites.

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Background: Malignant ascites (MA) is a common sequela of peritoneal carcinomatosis resulting in significant morbidity and poses substantial management challenges for clinicians. Large volume paracentesis remains the gold standard treatment. The impact of time to drainage of MA is poorly studied. The purpose of this study was to compare clinical outcomes of hospitalized patients with MA that had drainage performed within 48 hours of admission and after 48 hours of admission. **Methods:** The NIS was queried between the years 2016 and 2020 to identify all adult patients hospitalized with malignant ascites (ICD-10 code R18.0) and had a drainage procedure performed (ICD-10 Procedure code 0W9Gx). These patients were then stratified into an early cohort (<48 hours after admission) and late cohort (>48 hours after admission). Baseline characteristics and clinical outcomes were collected and analyzed using chi squared tests, independent sample t-tests, and binary logistic regression (adjusted for age, gender, and Charlson comorbidity index or CCI). Adjusted Odds ratios (aOR) are presented with 95% Confidence intervals (CI). **Results:** A total of 129,040 patients were analyzed, with 89,905 (69.7%) hospitalizations in the early cohort and 39,135 (30.3%) in the late cohort. Baseline characteristics were similar between the groups, with no significant differences in age (63.8 vs 63.6, p = .06) or gender distribution (63.6% vs 63.1%, p = .09). However, patients in the early paracentesis group exhibited a lower CCI (9.8 vs 10.2, p<.001). Notably, early paracentesis was associated with significantly improved outcomes, including lower mortality (aOR 0.67, 95% CI 0.64–0.69), reduced incidence of sepsis (aOR 0.67, 95% CI 0.65–0.69), decreased need for transfusions (aOR 0.69, 95% CI 0.66–0.71), lower rates of mechanical ventilation (aOR 0.44, 95% CI 0.42–0.47) and vasopressor use (aOR 0.79, 95% CI 0.73–0.87). **Conclusions:** Early paracentesis in patients with malignant ascites was found to be associated with significantly improved clinical outcomes, independent of comorbidity burden. These findings underscore the importance of timely intervention to mitigate complications and enhance quality of life.
Research Sponsor: None.

	Late Paracentesis N=39,135	Early N=89,905	p-value
Age (years)	63.6 ± 13.1	63.8 ± 13.0	.06
Gender (% female)	24665 (63.1%)	57120 (63.6%)	.09
CCI	10.2 ± 3.1	9.8 ± 3.2	**<.001**
Mortality	5720 (14.6%)	9030 (10.0%) aOR .67 (.64 - .69)	**<.001**
Sepsis	7675 (19.6%)	12525 (13.9%) aOR .67 (.65 - .69)	**<.001**
Transfusion	5275 (13.5%)	8605 (9.6%) aOR .69 (.66 - .71)	**<.001**
Ventilation	2130 (5.4%)	2190 (2.4%) aOR .44 (.42 - .47)	**<.001**
Vasopressor	770 (2.0%)	1385 (1.5%) aOR .79 (.73 - .87)	**<.001**

TPS843 **Trials in Progress Poster Session**

EQUITY GI: A prospective study to enhance quality, inclusivity, and trial participation in Black patients with gastrointestinal cancer.

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Background: Black individuals, including African Americans, experience a disproportionate cancer burden and exhibit the lowest survival rates among all racial groups for gastrointestinal (GI) cancers. The EQUITY GI study seeks to address the significant health disparities encountered by Black patients with GI cancers in the areas of biomarker testing, evidence-based care, clinical trial participation, and health literacy. Our central hypothesis postulates that a comprehensive approach—encompassing appropriate biomarker testing, ensuring evidence-based treatments, support for clinical trial participation, and health literacy initiatives—is essential to effectively mitigate health disparities among Black patients with GI cancers. **Methods:** The primary objective of this study is to promote equitable care for Black patients with GI cancers by ensuring appropriate biomarker testing and the delivery of evidence-based care. The secondary objectives include promoting clinical trial participation and enhancing health literacy. The specific goals are to increase the rate of evidence-based care, including biomarker testing, to ≥80%, increase the clinical trial enrollment rate from 5% to 15%, and improve the Cancer Health Literacy Test-30 (CHLT-30) score by 30%. Following a comprehensive assessment of the care gap, several initiatives will be implemented: 1. Tracking biomarker testing through a clinical information tracking tool (EQUITY-GI Oncotracker), 2. Providing guidance to treating providers through a molecular tumor board, 3. Facilitating referrals for clinical trials through navigators, and 4. Implementing health literacy interventions measured by CHLT-30 score. We aim to enroll 200 patients over 18 months. With this expected enrollment, the power to detect a ≥60% absolute improvement in the biomarker testing rate (from the current 20% to the expected ≥80%) is over 98%, using a two-sided chi-square test with a significance level of 0.05. To demonstrate an improvement in the clinical trial enrollment rate from 5% to 15%, 18 black patient will be required to get enrolled in a clinical trial through the referrals generated from the current project. To detect a 30% improvement in the mean CHLT-30 score (17 to 22) following health literacy interventions, at least 120 patients will be required, using a two-sided paired t-test with a significance level of 0.05. Research Sponsor: Merck.

TPS844 **Trials in Progress Poster Session**

A phase 1/2 study to evaluate CHM-2101, an autologous cadherin 17 (CDH17) chimeric antigen receptor (CAR) T cell therapy for the treatment of relapsed or refractory gastrointestinal cancers.

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Background: Patients with advanced gastrointestinal (GI) malignancies have poor prognoses and limited treatment options. Cadherin 17 (CDH17) is a cell membrane-associated protein important for GI cell-cell interactions that is expressed in proximity to epithelial tight junctions. In some GI malignancies, CDH17 expression is also expressed diffusely across the cancer cell membrane, thus providing access to CDH17 directed CAR T-cells. CHM CDH17 is a third generation autologous CAR T-cell product candidate that was designed to target and eradicate CDH17+ solid tumors. Notably, expression of CDH17 varies across indications in the cancer setting CDH17– TABLE 1. **Methods:** Clinical Trial NCT06055439 is a seamless Phase 1/2 clinical trial of autologous CHM CDH17 CAR T-cells for patients with advanced gastric cancer (GC), colorectal (CRC) cancer or neuroendocrine tumors (NETs) of the midgut or hindgut. As CDH17 expression in GC is heterogenous, GC subjects will be screened for CDH17 expression. In preparation for receiving CDH17 CAR T-cells, subjects receive 3-days of lymphodepleting intravenous (IV) chemotherapy (fludarabine and cyclophosphamide). The Phase 1 portion of the clinical trial uses a 3+3 dose escalation and a starting dose of 50 million CAR-T cells in a single IV infusion. Decisions regarding dose escalation/de-escalation are based on real-time dose-limiting toxicity evaluation during the first 28 days of treatment during dose escalation. After establishing a recommended Phase 2 dose, three indication-specific Simon 2-Stage cohorts will be initiated in (1) GC (2) CRC and (3) NETs of the midgut or hindgut to further characterize the safety and assess the efficacy of CHM CDH17 CAR T-cells. This is an ongoing first-in-human multi-center clinical trial of CHM CDH17, a CDH17 directed autologous CAR T-cell product candidate, enrolling subjects with advanced GI cancers that express CDH17. Clinical trial information: NCT06055439. Research Sponsor: Chimeric Therapeutics Ltd.

Expression of CDH17 in selected gastrointestinal cancer indications.

Indication	Sample size	CDH17 Expression %*
Colorectal cancer	821	96%
Neuroendocrine tumors of the midgut and hindgut	119	97%
Gastric cancer	1,409	64%

*Expression of total tumors tested for CDH17 by immunohistochemistry (CHM data on file).

TPS845 **Trials in Progress Poster Session**

MOONRAY-01, a phase 1 study of LY3962673, a potent, orally bioavailable, and selective KRAS G12D inhibitor in *KRAS G12D*-mutant solid tumors.

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Background: *KRAS G12D* mutations occur in ~35%, 13%, and 4% of pancreatic, colorectal, and non-small cell lung cancers, respectively and less commonly in other cancers. Currently, there are no approved therapies for patients with *KRAS G12D*-mutant solid tumors. LY3962673 is a potent and orally bioavailable inhibitor of GDP-bound *KRAS G12D* that is highly selective for *KRAS G12D* against wild-type KRAS and other RAS mutations or amplifications. Preclinically, LY3962673 demonstrated robust anti-tumor activity *in vivo* as a single-agent and in combination with other agents in multiple KRAS G12D PDX models (Gong X, et al. Poster presented at AACR 2024). **Methods:** MOONRAY-01 is a global, open-label, multicenter, first-in-human phase 1 study of oral LY3962673 in participants (pts) with *KRAS G12D*-mutant solid tumors conducted in 2 phases: phase 1a dose escalation (Part A) to evaluate LY3962673 monotherapy and phase 1b dose expansion/randomized dose optimization (Parts B-E) to evaluate LY3962673 monotherapy and in combination with other anticancer therapies (Table). In phase 1a, additional pts will be allowed to backfill to previously cleared dose levels that demonstrate therapeutically relevant exposures or direct evidence of clinical activity. In phase 1b, each combination cohort will include a safety lead-in of 3-6 pts. Phase 1a/b escalation will utilize the mTPI-2 method. The DLT evaluation period will be 28 days. Eligible pts (≥18 years) must have locally advanced, unresectable, and/or metastatic cancer, measurable disease per RECIST v1.1, evidence of *KRAS G12D* mutation in tumors or ctDNA, and an ECOG PS of 0-1; asymptomatic CNS disease is allowed. In phase 1a, pts must have received ≥1 prior line of systemic chemotherapy for advanced or metastatic disease. In phase 1b, prior KRAS therapy is not allowed; other eligibility requirements are outlined in the Table. Key exclusion criteria include known active CNS metastases and/or carcinomatous meningitis, or significant cardiovascular disease. Key objectives are to determine the RP2D/optimal dose and assess the safety, tolerability, PK properties, and antitumor activity of LY3962673 per RECIST v1.1. The study is currently enrolling pts. Clinical trial information: NCT06586515. Research Sponsor: Eli Lilly and Company.

Phase 1b Parts	Key Eligibility	Cohorts
B: PDAC	· B1: ≥1 prior systemic therapy · B2, B3, B4: treatment naïve allowed · B5: treatment naïve only	· *B1: LY3962673 · B2: LY3962673 + gemcitabine (G) + nab-paclitaxel (P) · *B3: LY3962673 + G + P · B4: LY3962673 + mFOLFIRINOX · *B5: LY3962673 + mFOLFIRINOX
C: CRC	· C1, C2: ≥1 prior systemic therapy · C3, C4, C5: treatment naïve allowed	· C1: LY3962673 + cetuximab (C) · *C2: LY3962673 + C · C3: LY3962673 + C + FOLFOX · C4: LY3962673 + C + FOLFIRI · *C5: LY3962673 + C + FOLFIRI / FOLFOX
D: NSCLC	· ≥1 prior systemic therapy	· *D1: LY3962673
E: Other solid tumors	· ≥1 prior systemic therapy	· E1: LY3962673

*Cohorts randomized to various LY3962673 doses.

TPS846 **Trials in Progress Poster Session**

Phase 1/2 study of the trophoblast cell surface antigen 2 (TROP2) antibody-drug conjugate (ADC) sacituzumab tirumotecan (sac-TMT) as monotherapy or combination therapy in gastrointestinal cancers: LIGHTBEAM-02A.

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Background: There is a significant need for new tolerable and effective therapeutic options for colorectal cancer (CRC), pancreatic ductal adenocarcinoma (PDAC), and biliary tract cancer (BTC) that can provide survival benefit. TROP2 is expressed broadly in several cancers, including CRC, PDAC, and BTC. Sac-TMT (formerly MK-2870/SKB264) is an ADC consisting of a humanized antihuman TROP2 monoclonal antibody, a linker, and a cytotoxic belotecan–derivative topoisomerase I inhibitor. The nonrandomized, open-label, phase 1/2 LIGHTBEAM-02A study (NCT06428409) is being conducted to examine the safety and efficacy of sac-TMT in gastrointestinal cancers. **Methods:** Approximately 130 patients will be assigned to 1 of 3 cohorts based on their disease. Patients in cohort 1 (CRC; $n \leq 50$) must have previously treated locally advanced or metastatic colorectal adenocarcinoma; patients in cohort 2 (PDAC; $n \cong 40$) must have previously treated locally advanced or metastatic PDAC; patients in cohort 3 (BTC; $n \cong 40$) must have previously treated locally advanced or metastatic BTC. Cohort 1 is included in a dose-escalation phase to determine the recommended phase 2 dose of sac-TMT when used in combination with fluorouracil and leucovorin and an efficacy phase. In the dose-escalation phase, patients will receive sac-TMT 3 mg/kg or 4 mg/kg intravenously (IV) every 2 weeks (Q2W) on days 1 and 15 of each 4-week cycle plus 2400 mg/m^2 fluorouracil IV infused over 46-48 hours Q2W and 400 mg/m^2 leucovorin IV Q2W; in the efficacy phase, patients will receive the recommended phase 2 dose of sac-TMT with the same chemotherapy regimen as the dose-escalation phase. Patients in cohorts 2 and 3 will receive sac-TMT 4 mg/kg IV monotherapy Q2W on days 1 and 15 of each 4-week cycle. The primary objectives are to evaluate safety and tolerability and objective response rate per RECIST v1.1 by blinded independent central review (BICR). Secondary objectives are to evaluate duration of response and progression-free survival per RECIST v1.1 by BICR and overall survival. Enrollment is ongoing. Clinical trial information: NCT06428409. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

TPS847 **Trials in Progress Poster Session**

A phase 1 dose escalation/expansion study of GSK5764227 (GSK'227), a B7-homolog 3 (B7-H3) protein targeted antibody-drug conjugate (ADC), in patients with advanced solid tumors, including gastrointestinal (GI) cancers.

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Background: B7-H3 is an immune checkpoint protein overexpressed in multiple solid tumors with limited expression in normal tissues. GSK'227 (HS-20093), a novel B7-H3-targeted ADC, is composed of a human anti–B7-H3 monoclonal antibody linked to a topoisomerase I inhibitor via a protease-cleavable linker, and has shown acceptable safety and promising antitumor activity in patients of Asian origin with advanced solid tumors (NCT05276609; NCT05830123). The current study will evaluate the safety, tolerability, efficacy, and pharmacokinetics (PK) of GSK'227 in patients with solid tumors, including GI cancers, in a global population. **Methods:** This two-part (dose-escalation [1a] and expansion [1b]) global, open-label, Phase I study (NCT06551142) will enroll ~260 patients, with enrollment currently ongoing. Key eligibility includes: aged ≥18 years, histologically confirmed advanced solid tumors, including those with colorectal cancer, esophageal squamous cell carcinoma, and pancreatic cancer, with measurable disease per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1), Eastern Cooperative Oncology Group (ECOG) performance status of 0–1, and no prior B7-H3 treatment. Eligible patients will receive intravenous GSK'227 every 3 weeks (Q3W) until progression, toxicity, loss to follow-up, or death. For Phase 1a, a Bayesian optimal interval design will be used to determine the maximum tolerated dose. Phase 1a primary endpoints are safety and tolerability, including incidences of adverse events (AEs) and serious AEs. Secondary endpoints include objective response rate (ORR), disease control rate, duration of response, immunogenicity, and PK. The Phase 1b primary endpoint is progression-free survival (extensive-stage small cell lung cancer cohort) or ORR (GI and other solid tumors cohort). Secondary endpoints include additional efficacy assessments, PK, safety and tolerability. For Phase 1a and 1b, efficacy will be assessed per RECIST v1.1, with imaging conducted Q6W from first dose then Q12W after 24 weeks. Safety follow-up will be conducted at 30, 60, and 90 (±7) days after the last dose. Safety, tolerability, and efficacy analyses will be conducted using descriptive statistics and, for efficacy analyses, point estimates with 2-sided 95% confidence intervals. This abstract was previously submitted to the European Society for Medical Oncology (ESMO) Immuno-Oncology Congress 2024 and is submitted on behalf of the original authors with their permission. Funding: GSK (Study 223054). Medical writing support was provided by Avalere Health, funded by GSK. Clinical trial information: NCT05830123. Research Sponsor: GSK.

INSIGHT: A phase 3, randomized, open-label study of ripretinib vs sunitinib in patients with advanced gastrointestinal stromal tumor previously treated with imatinib with *KIT* exon 11 + 17/18 mutations.

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Background: Gastrointestinal stromal tumor (GIST) is the most common gastrointestinal sarcoma, with approximately 80% of cases driven by *KIT* mutations. Most patients (pts) with advanced GIST experience disease progression following first-line treatment with imatinib due to *KIT* secondary resistance mutations occurring most commonly in the ATP-binding pocket (exons 13/14) and/or activation loop (exons 17/18). Sunitinib is approved as second-line therapy for advanced GIST. Ripretinib is a switch-control tyrosine kinase inhibitor approved for pts with GIST who received prior treatment with 3 or more kinase inhibitors, including imatinib. In the INTRIGUE phase 3 study (NCT03673501) in second-line therapy for advanced GIST, ripretinib was not superior to sunitinib in terms of progression-free survival (PFS); however, a more favorable safety profile was observed with ripretinib vs sunitinib (Bauer S et al. *J Clin Oncol*. 2022). Exploratory mutational analysis from INTRIGUE using baseline circulating tumor DNA (ctDNA) demonstrated that pts harboring primary *KIT* exon 11 mutations with secondary resistance mutations exclusively in *KIT* exons 17/18 derived PFS benefit with ripretinib vs sunitinib (median, 14.2 vs 1.5 months; HR, 0.22; 95% CI, 0.11 to 0.44; nominal $P < 0.0001$; Heinrich MC et al. *Nat Med*. 2024). Here, we describe an ongoing phase 3 study for pts with advanced GIST previously treated with imatinib harboring *KIT* exon 11 + 17/18 mutations. **Methods:** INSIGHT (NCT05734105) is a phase 3, randomized, open-label study designed to evaluate the efficacy of ripretinib vs sunitinib in pts with advanced GIST previously treated with imatinib and who harbor *KIT* exon 11 + 17/18 mutations. Eligible pts must be ≥18 years old with histologically confirmed GIST and co-occurring *KIT* exon 11 + 17/18 mutations confirmed by ctDNA analysis. Pts must also have advanced disease with ≥1 measurable lesion per modified Response Evaluation Criteria in Solid Tumors (mRECIST) v1.1, radiologic progression on imatinib, and an Eastern Cooperative Oncology Group performance status ≤2. Key exclusion criteria include a *KIT* exon 9, 13, or 14 mutation confirmed via ctDNA analysis at screening and prior treatment with another line of therapy in addition to imatinib for advanced GIST. A total of 54 pts will be randomized (2:1) to receive ripretinib 150 mg once daily (QD; continuous) or sunitinib 50 mg QD (4 weeks on/2 weeks off) in 6-week cycles. The primary endpoint is PFS by independent radiologic review (IRR) per mRECIST v1.1; key secondary endpoints are objective response rate by IRR using mRECIST v1.1 and overall survival. Safety and patient-reported outcome measures will also be assessed. Pts randomized to the sunitinib arm may cross over to the ripretinib arm upon disease progression. This abstract was originally presented at ASCO 2023. Clinical trial information: NCT05734105. Research Sponsor: Deciphera Pharmaceuticals, LLC.

TPS849 **Trials in Progress Poster Session**

A prospective multi-site translational study investigating the association of gut microbiome (GM) diversity with pathological complete response (pCR) after neo-adjuvant treatment in early stage rectal and esophageal cancer.

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Background: The diversity of the GM is defined as the number and relative abundance distribution of distinct types of microorganisms colonizing within the gut. Studies have suggested that dysbiosis of the GM confers a predisposition to certain malignancies and impacts response to immunomodulating therapies. The influence of the GM diversity on the pathological response after neoadjuvant chemotherapy and radiotherapy is unclear. Some studies have suggested that the GM may offer predictive biomarkers for response to chemo-radiation in rectal cancer. Other studies in early-stage rectal cancer patients indicated an association between GM diversity and pathological outcomes following neo-adjuvant therapy (NAT). We hypothesize that a more diverse GM constitution at baseline leads to an improved pathological response at the time of definitive surgery. **Methods:** We designed a cross-institutional translational study investigating the impact of the GM diversity on the efficacy of NAT in GI cancers by assessing its association with pathological response. The study population includes patients with early-stage rectal or esophageal cancer due to commence NAT (including chemotherapy and chemoradiation) who are planned for definitive surgery. Exclusion criteria includes prior allogenic tissue/solid organ transplantation and prior receipt of anti-cancer therapy. The study assessments include fecal sampling of the GM prior to NAT, upon completion and again six months post completion of therapy. Fecal samples are analysed by 16S RNA sequencing. Pathological response will be examined at time of surgery and patients will be classified as responders (complete pathological response) or non-responders. The primary endpoint of the study is to examine the association between the GM diversity and pathological response. Exploratory analysis will include the assessment of the association between cf-DNA and the GM diversity as well as an assessment of the association between cf-DNA at baseline and pCR. Species richness (Alpha Diversity) will be analysed using the Shannon diversity index and Jaccard similarity index will be used to calculate beta diversity. Following planned study recruitment, classification and clustering analysis will be performed with Principal Component Analysis (PCA) and Random Forest analysis. Logistic regression analysis adjusting for potential confounding factors will be employed to assess the primary endpoint of the association between GM and complete pCR in the final statistical analysis. Adjusted odds ratios (OR) and 95% confidence intervals will be presented. This trial accrued 11 patients between May 2023 and Sept 2024. Out of the 11 patients enrolled, 9 patients have undergone their planned surgery. We are expecting to have 30 patients accrued prior to Jan 2025. Research Sponsor: Science Foundation Ireland and Breakthrough Cancer Research.

TPS850 **Trials in Progress Poster Session**

Evaluating novel therapies and circulating tumor DNA (ctDNA) as a marker of response in curatively treated gastrointestinal cancers with microscopic residual disease.

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Background: ctDNA can identify the earliest sign of relapse (MRD) after curative therapies with specificities of 93-100%, positive predictive values over 95%, and median lead times of 8-9 months before radiographic relapse. ctDNA has identified a "new stage" of high-risk patients (pts) with no evidence of disease radiographically (rNED) and MRD who lack treatment (tx) strategies as, traditionally, they wait for radiographic progression of disease (POD) before starting therapies. Treating MRD is a promising strategy supported by observational studies where ctDNA clearance is linked with improved survival and on-tx kinetics predict drug efficacy. We designed a pilot investigator-initiated trial to 1) determine the feasibility of using ctDNA as a rapid signal for positive therapeutic activity and 2) obtain pilot data on the efficacy of a novel combination therapy (atezolizumab [A] + bevacizumab [B]) in eradicating MRD in previously curatively treated pts with GI cancers. **Methods:** We will enroll 20 pts with any-stage cancers divided over 4 cohorts (n=5 colorectal [CRC], n=5 hepatocellular/biliary tract carcinoma, n=5 upper GI, n=5 pancreatic adenocarcinoma) who have a positive Signatera ctDNA test and rNED any time after completing standard curative therapies. Pts are treated with A 1200 mg IV + B 15 mg/kg IV on day 1 of a 21-day cycle for up to 1 year with imaging every 12 weeks and serial ctDNA testing every 3 weeks. Co-primary endpoints are rates of pt enrollment in 12 months and rates of ctDNA responses at a 12-week landmark after tx initiation. Pts with ctDNA POD (ctDNA doubling on each of 2 sequential tests, any rate of rise in 3 sequential tests, or radiographic relapse) will stop while those with ctDNA complete response (CR; clearance on 2 sequential tests + ongoing rNED) or ctDNA partial/stable response (PR; not ctDNA CR or POD + ongoing rNED) will continue tx. Each cohort's Bayesian predictive probability of positive therapeutic activity will be calculated. If at least 60% of a cohort experiences ctDNA CR, it will be expanded by 5 pts. This scenario is equivalent to Simon's 2-stage Minimax design testing a null hypothesis of 20% CR vs. an alternative hypothesis of at least 60% CR at 5% alpha with 80% power. If at least 5/10 pts in an expanded cohort experience ctDNA CR, this will generate interest in developing a larger confirmatory trial to evaluate pt survival outcomes with A+B and ctDNA as a surrogate for survival. Secondary endpoints are toxicity and enrollment barriers. Exploratory endpoints are associations between ctDNA conversion and disease-free survival, tumor whole exome sequencing data, and peripheral blood immune profiles collected at baseline and on tx. Enrollment began in Q1 2023, paused in 7/2023 due to funding changes, and re-opened 8/2024 with 5 patients enrolled currently (n=4 CRC, n=1 biliary). Clinical trial information: NCT05482516. Research Sponsor: Genentech, Inc; Natera, Inc.